



JOMO KENYATTA UNIVERSITY
OF
AGRICULTURE AND TECHNOLOGY
BSc. MARINE ENGINEERING
PROJECT REPORT
DESIGN, FABRICATION AND TESTING OF A LOCALLY
REMOTE OPERATED VEHICLE FOR SHIP HULL
INSPECTION

PROJECT CODE: MR-PRO-24-04

STUDENTS:	DENNIS SAMMY	ENM 241-0117/2019
	PEGGYOLANDA NYAMBURA	ENM 241-0118/2019
SUPERVISOR:	ENG. ELIJAH MUNYAO	

This project report has been submitted in partial fulfillment of the requirements for the award of degree of Bachelor of Science Marine Engineering

DECLARATION

We declare that this project is our original work. To the best of our knowledge, we also affirm that this project has not been presented to this university or any other university for examination, or for any other purpose.

Name: **DENNIS SAMMY** **ENM 241-0117/2019**

Signature: _____ Date _____

Name: **PEGGYOLANDA NYAMBURA** **ENM 241-0118/2019**

Signature: _____ Date _____

CERTIFICATION

I certify that the above-mentioned students carried out the work detailed in this project under my supervision.

Supervisor's Name: **ENG. ELIJAH MUNYAO**

Signature: _____ Date _____

Abstract

Hull inspection is an essential process for any sea going vessel to ensure safety of navigation. Over the years, hull inspection has been achieved through dry docking as well as deploying inspectors to carry out underwater inspections. However, this has had limitations due to the inadequate human capabilities underwater as well as the associated safety risks. Remotely operated vehicles (ROV) eliminate the risks that divers are subjected to during hull inspections and minimises expenditure on the entire hull inspection process in dry docks. The use of an ROV as an underwater inspection tool provides precise and efficient underwater inspections at reduced risks, associated costs, and docking time periods. The current work involved the design, fabrication, and testing of the remotely operated vehicle focusing on the development and implementation of the control system optimized for underwater hull inspections. The ROV integrates structural, electrical, and control systems to ensure precision, reliability, and adaptability in submerged conditions. The frame, constructed using acrylic, is engineered to withstand pressures at depths of up to 100 meters, verified through Finite Element Analysis (FEA). Acrylic underwent a maximum stress deflection of 57.394 Mpa and A maximum deformation of 2.083mm at the depth of 100m. This lies within the allowable stress and deformation limits. A dual battery system power five brushless thrusters. The control system, powered by a Raspberry Pi and operated via a joystick controller, enables real-time control. Sensor integration, including an MPU6050 for orientation data, ensured precise stabilization.

Table of Contents

Abstract	iv
List of Figures	vii
List of Tables	viii
1. Introduction	1
1.1 Background	1
1.2 Problem statement	2
1.3 Objectives	2
1.3.1 Specific objectives	2
1.4 Scope	2
2. Literature review	3
2.1 Work Class ROV	4
2.2 Observation Class ROV	5
2.3 Micro ROV	5
2.4 Work Class ROV	5
2.5 Special Use ROVs	5
3.0 Methodology	7
3.1 Conceptual Design	7
3.1.1 Structural design	7
Design constraints	7
Hydrostatic and hydrodynamic considerations	8
Design Considerations and requirements	8
Pressure Calculation	9
Selection of materials	9
Analysis of proposed materials	11
3.1.2 Propulsion mechanism Design	11
Thrusters	11

Thruster configuration	12
Analysis of upthrust, buoyancy, drag force, power of the motor and the thrust Force.	12
3.1.3 Electrical and control design	14
3.2 Fabrication and assembly	21
3.3 Testing	24
4. Results and Discussion	25
4.1 Research and selection of potential materials for key components.	25
4.2 Analysis of proposed materials	25
4.3 Determining the weight and displacement of the ROV.	37
4.4 Calculating the buoyant force, hydrodynamic drag force, power of the motor, and the thrust Force.	39
4.4.1 ROV Upthrust Force	39
4.4.2 ROV buoyancy	39
4.4.3 ROV hydrodynamic drag	39
4.5 The control system functional implementation	42
4.6 Propeller	44
5 Conclusion	45
6. Budget	46
7. Time plan	47
References	48
Appendix	50
2. Thruster Service code	51
3. Video service code	54
4. Sensor Service code	55
5. Propellor schematics	57
	57

List of Figures

Figure 1: Rebikoff-niggeler Foundation First ROV	11
Figure 2: ROV explores the ocean depth.	12
Figure 3: Electronic System Architecture	24
Figure 4: High Level system architecture.	24
Figure 5: Li Po battery pack with BMS	25
Figure 6: Raspberry pi 4	26
Figure 7: Electronic speed controller	27
Figure 8: Brushless motor with propellor attached	27
Figure 9: Mpu6050: Inertial Measurement Units	28
Figure 10: An ethernet cable used for transmission of data	28
Figure 11: Frame model	34
Figure 12: Von Mises Stress and Deformation Using Steel	34
Figure 13: Von Mises Stress and Deformation Using aluminium	35
Figure 14: Von Mises Stress and Deformation using PVC.	35
Figure 15: Von Mises Stress and Deformation Using Acrylic	35
Figure 16: Stress comparison at a depth of 10 m	36
Figure 17: Deformation comparison at a depth of 10 m	36
Figure 18: Deformation comparison of each material at different depths	43
Figure 19: Stress comparison of each material at different depths	43
Figure 20: Modelled acrylic sealing compartment	44
Figure 21: Stress and Deformation of Acrylic sealing compartment at 100m depth.	44
Figure 22: End Caps	45
Figure 23: Screenshot of the GUI showing controls for movement and status indicators.	51

List of Tables

Table 1:Pros and cons of proposed material	18
Table 2: Technical characteristics of existing thruster model	21
Table 3: Stress and Deformation of Steel on Depth Variation	38
Table 4:Stress and Deformation of Steel on Depth Variation - values	38
Table 5: Stress and Deformation of Aluminium on Depth Variation	39
Table 6: Stress and Deformation of aluminium on Depth Variation - values	39
Table 7: Stress and Deformation of PVC on depth Variation:	40
Table 8: Stress and Deformation of PVC depth variation - values	40
Table 9: Stress and Deformation of Acrylic on depth Variation	41
Table 10: Stress and Deformation of Acrylic on depth variation - values	41
Table 11: Properties of materials	42
Table 12: Allowable stress for the different materials	42
Table 13: Stress and deformation of acrylic on depth variation - values	44
Table 14: Total Mass of the ROV	45
Table 15: Performance metrics of control system	50
Table 16: Budget	55
Table 17: Time Plan	56

1. Introduction

1.1 Background

In the shipping industry, ships are required to be inspected over a given period as safety measures [1]. Currently, there are significant challenges associated with conducting hull inspections. These include the high cost, a time-consuming process involved in inspection of hulls in dry docks. In cases where underwater hull inspections are adopted, the inspectors are exposed to a myriad of safety risks as well as the inability to access some areas of the submerged hull. The hull is exposed to underwater processes such as biofouling and corrosion, making its inspection vital and therefore it should be conducted regularly, which is not feasible with the currently available method. Both Biofouling and corrosion have a great negative impact on the performance, condition and lifespan of the ship. Biofouling has also been identified as a contributor to increased fuel consumption for sea going vessels, consequently contributing to increased emission ozone depleting gasses - the greenhouse gasses [2].

Currently, the existing hull inspection methods include dry docking and use of underwater hull inspectors [3]. In dry docking, a ship is unsubmerged for a comprehensive inspection to be conducted on the hull, the rudder, propeller, and stabilizers. Dry docking has proven to be expensive and time consuming considering the required labour, resources and planning. The docking facilities are also not readily available resulting to delays in inspection. Another alternative involves utilizing underwater inspectors to evaluate the status of the hull. Even though this reduces the amount of resources and cost expended in the inspection, it has proven to be a difficult and risky endeavour. It is life threatening as it exposes the inspectors to hydrostatic pressures, low visibilities, marine predators, rough and harsh sea conditions [4]. Due to these limitations, there is a need to develop alternative reliable and safe methods to conduct hull inspection. One of the feasible approaches is to use a Remotely operated vehicle (ROV).

A Remotely operated vehicle (ROV) is a manoeuvrable underwater machine that can be used to explore and inspect ocean depths operated by personnel at the water surface. ROVs are versatile tools that find applications in oil and gas companies, marine construction works, military and law enforcements applications. They are also used in underwater research as well as in shipping industry, bringing forth numerous benefits in these areas of application [5]. ROVs allow the accessibility of inaccessible underwater areas. They also guarantee improved safety and operate with low maintenance. They have been used to recover torpedoes, bombs, and mines from the 1960s by the navy as well as to carry out missions that are very dangerous to humans such as mine breaking and

mineral harvesting [6]. ROVs have also been used by the leading coating provider, Hempel, to perform remote hull inspections [7]. It is based on the above discussion that this project implements a locally made ROV for hull inspections.

1.2 Problem statement

Inspection of ship hulls and offshore structures is a major challenge in the maritime sector. Conventional hull inspection methods such as docking, and use of underwater inspectors have proven to be costly and risky. An alternative way to overcome these challenges is to create a device which will fulfill the underwater duties. Remotely Operated Vehicles (ROV's) have become vital tools in industries that are related to underwater. Deployment of ROVs has been proven to reduce the inspection cost from the previously existing methods but still costly for the local people. The existing ROVs are relatively expensive due to the intensive research and high-end technology put into their development, and this is a prohibitive barrier to their entry into application, preventing widespread adoption of ROV for personal, research, and conservation applications. The current work therefore seeks to address the challenge by coming up with a low cost locally available Remotely Operated Vehicle without compromising its functionality.

1.3 Objectives

The main objective of the current work was to design, fabricate and test a local Remotely Operated Vehicle (ROV) for hull inspection.

1.3.1 Specific objectives

1. To design and test the structural integrity of the ROV frame.
2. To design, fabricate and assemble the control system of the ROV.
3. To test the responsiveness, manoeuvrable controls, and stability of the control system of the ROV.

1.4 Scope

The scope of this project is to design, fabricate and test the control system of the Remotely Operated Vehicle. The ROV should maintain structural integrity as well as display manoeuvrable controls along the main axes of motion and around an underwater obstacle.

2. Literature review

Over time, humans have creatively built tools to improve lives and the ROV is no different. The human desire to reach places underwater that are too dangerous or too deep for a diver to venture into has seen the development of ROVs. ROVs are unoccupied machines designed for specific underwater missions. Typical ROVs are controlled from a vessel, land, or a floating platform. The first tethered ROV referred to as POODLE was developed by Dimitri Rebikoff in 1953-1954. The United States navy later advanced the ROV technology to recover bombs, mines, and torpedoes in the 1960s that had been lost during tests. ROVs gained popularity when the Cable Controlled Underwater Recovery Vehicle systems were used by the US Navy to recover an atomic bomb lost off Palomares Spain and then saved pilots of a sunken submersible off Cork in Ireland with only minutes of air remaining [8].

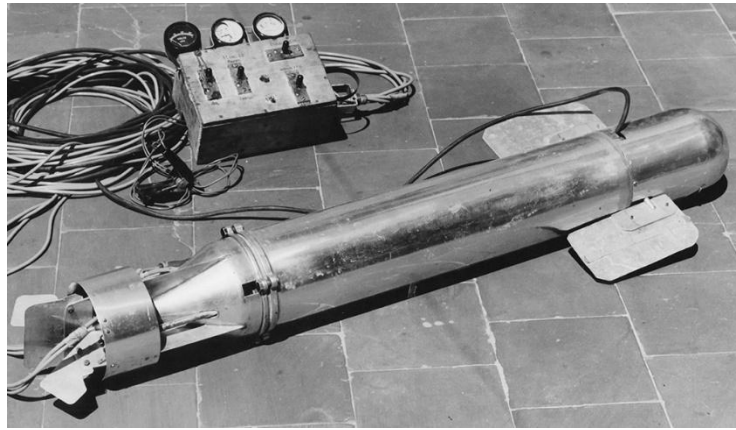


Figure 1: Rebikoff-niggeler Foundation First ROV

The ROVs were also used on rescue missions that exceeded the reach of human divers such as mine hunting and mine breaking. Since the 1980s, the offshore oil industry has successfully used ROVs to assist in offshore development, especially as waters became too deep for human divers. ROVs have become famous for discovering historic shipwrecks, including the famous RMS Titanic in 1985 and the Bismarck [9]. The first ROVs were initially referred to as Cable-Controlled Underwater Recovery Vehicles (CURV) since their primary function was to lead search and rescue operations in deep and dangerous waters.

ROVs consist of video cameras that transmit real time surveillance to scientists aboard vessels and this should be with low latency, sonar systems and a buoyancy pack that allow the vehicle to remain light and easy to manoeuvre when in the water. Thrusters are used to manoeuvre the vehicle, lights to provide illumination for the camera underwater, a frame that provides a structure to attach the ROV components, and a tether which carries electrical power and signals to the surface. ROVs use

external sensors that are mounted on the vehicle to measure properties or parameters such as temperatures and depth. ROVs may be built with end effector such manipulators for collecting biological and geological samples.

Some ROVs are built with two bodies such as the NOAA Ocean Exploration' vehicles [10]. The advantage of a two-body system is that the hovering ROV acts as an extra light source and camera giving pilots, scientists, and viewers an expanded view of the ocean. However, this has resulted in difficulties in control and transportation of single body counterparts. ROVs range in size, likewise the prices also vary a great vast too depending on the size and application of the ROV. They normally start at thousands of dollars and can go up to millions of dollars. The figure below shows an ROV using LED lights to explore the ocean floor.



Figure 2: ROV explores the ocean depth.

Among the prominent ROVs is ROV Ventana. ROV Ventana was constructed by International Submarine Engineering in 1987 and launched by MBARI's research vessel and has been in operation since 1988. It runs on hydraulic power, exploring waters at a depth of 1800 meters [11].

Underwater ROVs can be classified into various classes based on their weight, power, abilities and sizes. The common classes include the work class ROV, light work class, observation class ROV, and the micro ROV.

2.1 Work Class ROV

This class of ROV is used for ocean floor exploration and inspections at depths that divers are often unable to reach. Oil and gas exploration and production activities in the marine environment are increasing. In addition, the amount of global hydrocarbon reserves removed from below the seafloor is set to increase, with exploration heading into deeper, more remote waters, many of which are yet

to be fully explored [12]. Work Class ROVs act as a safe alternative to divers and are often used in offshore energy projects and deep archaeological explorations.

2.2 Observation Class ROV

This class of ROV is small and is used to explore lakes, rivers, and coastal waters. They are often used to test water safety prior to a diver entering the water during missions and conduction inspections. They are equipped with sonar and custom sensors that make them versatile underwater vehicles.

2.3 Micro ROV

The ROVs in this class are the smallest and are often used to inspect hard to reach areas at shallow depths such as pipe systems and submerged infrastructure. They explore miniscule cavities and pipeline cracks which are physically impossible for divers to achieve.

2.4 Work Class ROV

This class of ROV is used for ocean floor exploration and inspections at depths that divers are often unable to reach. They act as a safe alternative to divers and are often used in offshore energy projects and deep archaeological investigations.

2.5 Special Use ROVs

They describe tethered underwater vehicles designed for specific purposes. An example of a special-use vehicle is a cable burial ROV system designed to plow the sea floor to bury telecommunications cables.

ROVs have great benefits that enable them to be a prospective candidate in providing a solution to underwater exploration, inspections, and adventures. They have been able to have quick deployment due to the compact design and easy-to-use inspection technology. This is highly beneficial in emergency situations and in areas that are narrow or difficult to access. ROVs are robust and rugged in design thus can withstand harsh water environments and require minimal maintenance. Depending on sea conditions and the type of operations, divers can only remain underwater for a limited amount of time; 30 min to an hour at a time. ROVs have overcome the challenge as they are designed to operate for long hours and in difficult conditions. One of the most remarkable benefits is the photo and video recording capabilities. ROVs can provide high resolution footage even in dark and murky waters. ROVs can be deployed to record and later be reviewed for documentation and results. In addition, ROVs provide safe alternatives for exploration and cost effectiveness comparable to traditional methods [13].

Several industries have been able to benefit from the use of underwater ROVs technology. These industries include Military, Aquaculture, Municipal Infrastructure, Commercial and Salvage Diving, Ocean Science, Oil and energy, Shipping, Underwater Discovery.

3.0 Methodology

The current work was carried out systematically in three main steps: conceptual design, fabrication, and testing.

3.1 Conceptual Design

Conceptual design comprises the design of structural, electrical and control systems. The design aims to reduce the cost of the ROV by considering locally available materials and technologies without compromising the functionality of the ROV.

3.1.1 Structural design

Structural design is an essential key part of the project as it assists in coming up with a sturdy and robust structure for the ROV. This design entailed:

- Determining the main dimensions; length, width, height.
- Determining weight of the ROV.

This provides the basis to create a 3D model that is essential in the successive sections of the design. A key part of the structural design is the selection of materials that are used to form the structure to withstand the high hydrostatic and hydrodynamic pressures that the ROV will be subjected to during its operation.

Design constraints

The major constraints that affect the design process of the Remotely Operated Vehicle include:

a) Environmental constraints

The vehicle is to operate in sea waters, thus the parameters experienced are depth, density and salinity.

b) Density and salinity

The density of water is affected by temperature and salinity. Water is at its highest density at 4 degrees Celsius and the density decreases with an increase in saline concentration.

c) Working depth

The working depth defines the mechanical strength of all structural components. The current work suggests an operational depth of 10 m. This depth establishes a secure margin for the collapse resistance of the structural elements. Thus, the structural calculations are done at the design depth to guarantee the integrity of the ROV.

Hydrostatic and hydrodynamic considerations

A list of hydrostatic and hydrodynamic considerations is addressed by Allmendinger [14]. Such considerations include loads of interest, static equilibrium conditions, static and dynamic stability conditions and drag and thrust considerations.

- Load of interest

The loads that affect the ROV include weights, displacement force due to change in pressure, and drag and lift force as a result from the motion of the ROV in water,

- Static equilibrium

The buoyancy of the ROV is the difference between the displacement and its weight. The buoyancy is positive when the displacement is greater than weight and it is negative when the displacement is less than the weight. It is neutral when the displacement and the weight are equal. It is desired that the buoyancy be positive or neutral. This is achieved when the difference between the displacement and vehicle's weight be small and positive [15]

- Static and dynamic stability conditions

Static stability occurs when the heeling moment increases very slowly from zero to a maximum value while dynamic stability is when the heeling moment changes rapidly. Static stability is the most important measure of heeling behaviour of a ship. A vessel submerged in water will always seek to assume a position in which all forces and moments acting upon it cancel each other out, which means that it is in equilibrium when its weight and buoyancy are equal but of opposite directions [16]

- Drag and thrust.

The thrust power must be able to overcome drag at an operational speed, and during acceleration intervals.

Design Considerations and requirements

The main parts of the ROV include the frame, the sealing compartment, thrusters, camera, lighting equipment, control mechanism and buoyancy mechanism. The frame plays an important role of support and protection of the equipment [17]. It is thus necessary the following factors should be considered in the design of the ROV:

- Withstand pressure at the designed depth.
- Cost effective.

- Lightweight to ensure a neutral or negative buoyancy
- Water corrosion resistance.
- Material Machinability.
- Robust and Sturdy.

Pressure Calculation

There have been previous attempts to build the ROVs as documented in [18]. With the desired operational depth of 10m, pressure can be calculated to determine whether the selected structural material for the frame is suitable.

$$\rho \times g \times h \quad 1$$

Where P is the pressure, ρ is the density of water in Kg/m^3 , g is the gravity and h is the depth in meters. Taking ρ as density of sea water to be 1025 kg/m^3 , pressure is calculated using equation [1]. Where P is hydrostatic pressure in MPa, h is 10m, g is 9.81 m/s^2 and density of seawater is 1025 kg/m^3

$$10 \times 9.81 \times 1025 = 100552.5 \text{ Pa} \quad 2$$

Selection of materials

The frame of the ROV, the sealing compartments and the end caps are the three key components of focus in material selection.

• The Frame

The frame is the structural architecture that will hold the components that will drive and execute the functions of the ROV. The materials proposed include aluminium, Stainless Steel, unplasticized PVC (Polyvinyl Chloride), and Acrylic- Plexiglass [19]. The pros and cons of each material are shown in table 1 below.

Table 1: Pros and cons of proposed material

Material	Pros	Cons
PVC	Light in weight Excellent corrosion resistance Cost effective. Readily available. Durable	Susceptible to UV degradation Lower strength in comparison to metals
Aluminium	Light in weight Easy to machine and fabricate. Good corrosion resistance	Requires proper anodizing
Acrylic Plexiglass	Corrosion resistant Light in weight	Expensive
Stainless steel	High strength and durable Good corrosion resistance	Heavy Expensive

- **The Sealing compartment**

The sealing compartment is used to protect and house the electronic components of the ROV. The material used to make the sealing compartment of electronic components in Remotely Operated Vehicles (ROVs) is critical for ensuring the integrity and functionality of the electronics in underwater environments. Acrylic plexiglass can withstand the pressure at the designed depth. In addition, Acrylic offers excellent light transmittance and UV resistance. This is beneficial for visibility and protection in marine environments [20].

- **The End Caps**

End caps are used to enclose the openings of the cylindrical sealing compartments. They are to be machined using Teflon, which is commonly used on bearings and low-friction bushings due to its low friction, thermal resistance, and corrosion resistance [21]. The design consists of 4 O-rings on the end caps as well as a wide flange nitrile gasket on the face of the dome and the fixed dome connecting the aluminium mount. A nylon ring is machined so that it presses the transparent dome against the front cap once sealed with silicone so that no water enters.

Analysis of proposed materials

Selection of the best material for the frame entails subjecting the proposed materials to a stress and deformation analysis test under load. The modelled design is subjected through a series of pressures at different depths of 10m, 50m, and 100m and the results are reviewed. Fusion360 is used to analyse the structure for stress and deformation. The mechanical properties, thickness, and allowable values of each material are then summarized.

3.1.2 Propulsion mechanism Design

The propulsion system allows the ROV to utilize underwater thrusters in a specific configuration to grant an operational range of motion converting input energy into useful thrust. The propulsion system consists of motors and propellers - a combination called a thruster. The size of the propeller is described by a pitch and diameter number. Rake is the degree that the blades of a propeller slant forward or backwards in relation to the hub which is the centre section of the propeller. The aim of the design of a ROV propulsion system is to have high thrust to physical size and power input ratios [22].

A propeller is selected with consideration for performance, speed, weight and the overall size of the ROV. Ducted propellers move more liquid, and they increase the efficiency depending on the flow rate [23]. The selected propeller features three blades, a small diameter, a high rake angle and nylon polymer construction. Three bladed propellers are chosen because of the size of the ROV and the overall cost of the propeller. The smaller diameter propeller helps in avoiding collision with any of the frame components and to provide maximum thrust capabilities from the thrusters selected. A high rake angle allows for faster acceleration at lower rotations per minute thus allowing the ROV to respond rapidly to changes and reposition [24]. Due to weight and cost restrictions, nylon plastic propellers are selected to ensure the best performance to cost ratio.

Thrusters

As the propeller rotates in ROV, high-pressure and low-pressure areas are formed between the blades. The thrust needed for the movement of the ROV is formed as the fluid moves from the high-pressure area to the low-pressure area. The design of the thrusters ought to allow for components to be replaced in future in case of a component upgrade or a part failure. The individual thrusters are to have identical propellers to allow for each propeller to have uniform performance during operation [25]. This allows the operator to have more uniform control of the ROV despite external water conditions. As the propeller moves, there are losses at the tip of the propeller blades. To minimize the losses, a duct is designed around the propeller. The duct is designed with an aerodynamic structure to increase the efficiency of the design.

Brushless motor thrusters are chosen for the ROV since they are efficient and durable. The diameter of the propeller is chosen to be bigger than the motor diameter for enhanced thrust efficiency, reduced cavitation, lower power consumption, improved control and stability, optimized flow characteristics and hydrodynamic performance. The pitch of the blades depends on the diameter and the rotational speed of the motor in RPMs. The width of the blade determines the amount of water it pushes; thus light blades are used for higher speed applications [26]. They are designed to produce forward and reverse thrust. Each thruster is controlled using an electronic speed controller (ESC).

Thruster configuration

The ROV is designed with five thrusters. The thrusters mounted involve two thrusters moving in the horizontal motion along with three vertical thrusters. Two thrusters are placed on the back of the ROV equidistant from the centre of momentum in the X-direction to minimize rotation. They provide translational motion in the X-direction. The remaining three thrusters are placed on the top of the ROV which provides the motion in the Z-direction. They are placed along the X-axis for the ROV to obtain pitch rotation by only firing the front thruster for pitch down and firing only the back thruster for pitch up. This configuration allows to actively control the system in six degrees of freedom. The basis of this ROV system consists of 5 Blue Robotics T100 thrusters, which have the following technical characteristics depicted in table 2 below.

Table 2: Technical characteristics of existing thruster model

Property	Description
Thrust	14.7 N forward and 11.77N reverse
Dimensions	72 by 92 mm
Mass	0.075kg
Electric Power	Up to 150W when powered from 12V DC
Supply voltage	12-24V

Analysis of upthrust, buoyancy, drag force, power of the motor and the thrust Force.

Calculation of Upthrust Force:

Upthrust is the force exerted by a fluid on an object in an upward direction. It is equivalent to the weight of the fluid displaced by the submerged part of the body. Upthrust is calculated using equation (3) below such that m is the mass of the body and g is the acceleration due to gravity.

$$U = mg$$

3

Calculation of buoyant force.

The buoyant fore is the upward force exerted by the water on the ROV, which counteracts its weight. It is calculated using equation (4) below

$$Fb = -\rho \times g \times v$$

4

where p is the pressure, g is the gravity and v is the volume of the ROV.

The ballast weights and syntactic foam are either added or subtracted to fine tune the ROV's buoyancy and achieve a neutral buoyancy.

Calculation of Drag force.

The thrust required to achieve a design speed of 2m/s [27] depends on the hydrodynamic drag force acting on the ROV. To maximize the resulting velocity from a given propulsive force, the ROV ought to displace the least amount of water possible along its movement [28]. Thus, there should be an absolute minimum of surface area that obstructs water flow through the ROV yet retain structural integrity to withstand the propulsive forces without any deformation The drag force on an ROV moving through water can be calculated using the drag equation (5).

$$F_{D(body)} = \frac{1}{2} \times \rho \times v^2 \times C_d \times A$$

5

Where:

ρ is the density of sea water= 1025kg/m³ and gravitational acceleration = 9.8m/s²

v is the desired velocity (2 m/s)

C_d is the drag coefficient

A is the characteristic cross-sectional area of the ROV

The total drag of the ROV system is the vehicle drag, and the tether drag. The coefficient of drag for tether cables used in Remotely Operated Vehicle (ROV) applications vary depending on the cable's surface characteristics, diameter, and flow conditions. The tether drag coefficient in a turbulent flow regime is approximated to be 1.2 [29].

$$F_{D(tether)} = \frac{1}{2} \times \rho \times v^2 \times C_d \times A \quad 6$$

The total hydrodynamic drag force on the ROV system is thus the summation of the tether drag, and the vehicle drag, and it is given by equation (7) below.

$$F_{D(Total)} = F_{D(body)} + F_{D(tether)} \quad 7$$

Power of motor analysis

To calculate the power of the motor needed to achieve a speed of 2m/s and overcome the drag force experienced by the ROV, the fundamental relationship between power, force and velocity is applied in equation [7] below:

$$P_{req} = F_D \times V_{max} \quad 8$$

The thrust force, F_T , to achieve a desired speed is given by equation [8] below:

$$F_T = \frac{Power}{Velocity} \quad 9$$

Where P is Power in watts, F is force in newtons, v is velocity in m/s,

3.1.3 Electrical and control design

The electronic and control circuitry design of a Remotely Operated Vehicle (ROV) enables precise operation, communication, and functionality of the ROV in underwater environments. This design includes the integration of micro coprocessor, sensors, and communication modules to process operator commands and relay real-time data. Control circuitry governs the operation of thrusters ensuring smooth and accurate movements.

Control System Architecture

Figure 3 shows the block diagram of the ROV system architecture layout that consists of two major subsystems: the ROV system and the operator system. Figure 4 shows the high-level communication system architecture of the ROV. The two subsystems communicate through serial communication via an Ethernet Cable so that the user can send control commands to the ROV from a personal computer, laptop. The navigation control consists of buttons series interconnected by an Ethernet network cable to onboard computer, from which the user controls the ROV through a Graphical User Interface (GUI). The ROV subsystem includes a combination of services to manage the operation and execution of tasks. These systems include:

1. Communication service

The communication service handles relaying the commands sent by the operators to the respective services. It also manages the execution of parallel tasks to give an output. The Raspberry Pi 4 is at the centre of the subsystem and relays the processes to individual services as well as support a consistent communication with the operator interface. A serial socket communication protocol is used to enable the services support an active real time communication to reduce latency.

2. Thruster management service

This service includes the motors, ESCs and service control program. The control program receives directional and speed commands from the operator and assigns the right signals to the brushless motors through the ESCs. The control program also handles the management of the direction and speed conversion from the operator to signals.

3. Sensors management service

This service includes the IMU sensor that will be used to measure and record the different payloads necessary and crucial during operation. The IMU enables the measurement of pressure, temperature, gyroscopic values that is yaw, roll and pitch as well as the accelerometer. The feedback is relayed back to the operator interface.

4. Visual management service

This service manages the camera and visual feed from the camera module. The service fetches image frames and converts the to binary and generates a buffer stream to enable the transmission of data over serial connection.

5. Power Management Service

This subsystem handles delivering power to the separate services that are needed. Power intensive services such as the Thruster service uses power distributed by the power services.

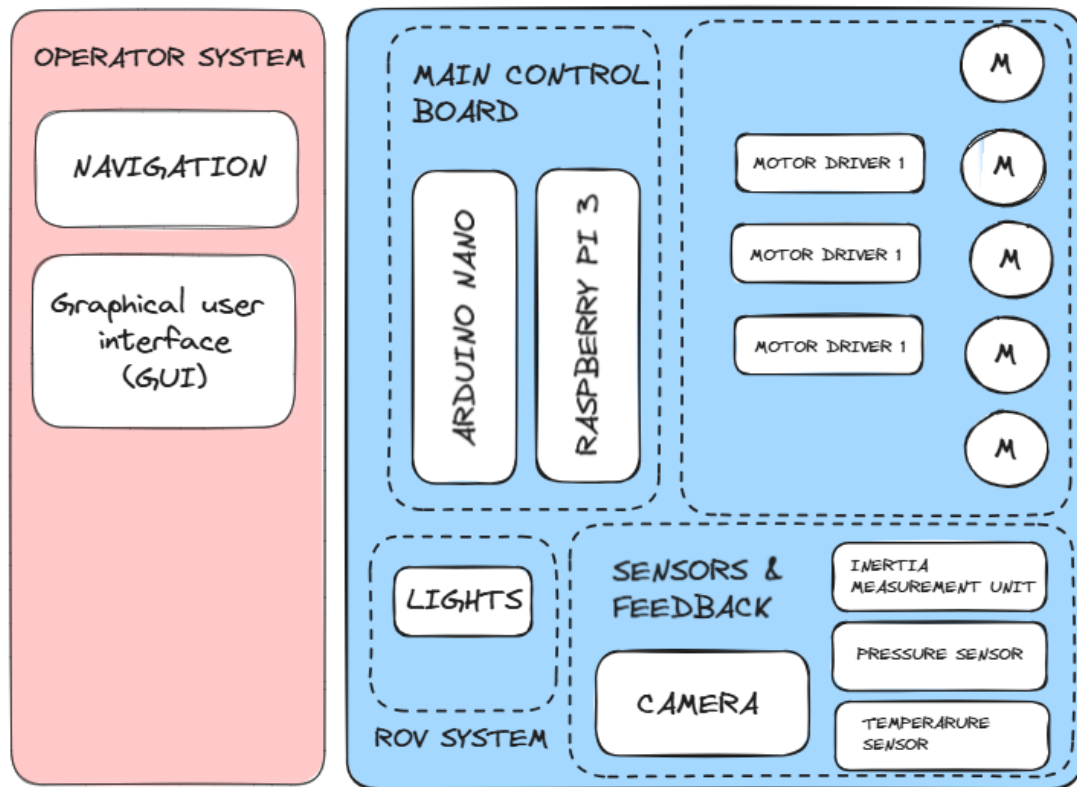


Figure 3:Electronic System Architecture

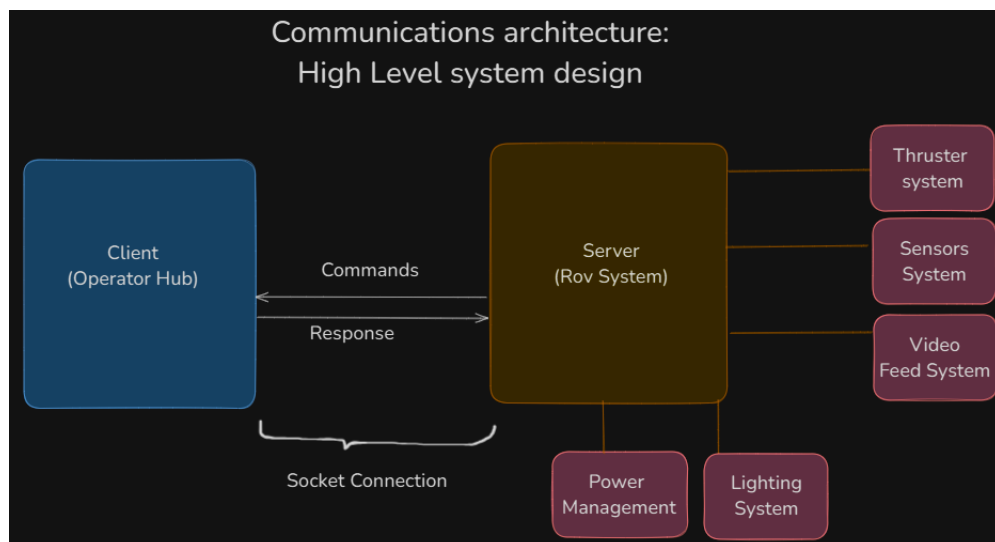


Figure 4: High Level system architecture.

Electrical components

The electrical devices of the Remotely Operated Vehicle (ROV) include key components that enable its functionality, control, and communication in underwater environments. These devices typically consist of thrusters for propulsion, cameras for visual feedback, and sensors to measure parameters

such as depth, temperature, and water pressure. Raspberry pi 4 is used to process commands and control the ROV's.

1. Power system

The power system describes how the ROV components are powered. This project used batteries as the power source. The batteries are to be reusable hence rechargeable Li-Po batteries are used. Due to the presence of motors that are used to power the thrusters more power is drawn from the batteries. A proposed solution is to allocate a second power source to provide power to the motors when needed. Figure 5 below shows the Li-Po battery pack with the Battery Management System (BMS).



Figure 5: Li-Po battery pack with BMS

2. Control system

The control system manages the input from the operator controller at the surface and converts it into control motion of the ROV underwater. Measurable parameters such as voltage, battery life, depth, and position of the ROV are recorded by sensors and displayed to the operator.

3. Microprocessor

The microprocessor is responsible for executing parallel tasks such as motor control, measurement and recording of different sensor reading of the system, battery voltage monitoring for energy consumption and communication management with remote control. The Raspberry PI 4 is the selected microprocessor. It is a compact, affordable, powerful and versatile single board computer (SBC) that serves for the control system of the ROV. A camera module connects to the board of the

Raspberry pi 4 via the Camera Serial Interface (CSI) port using a flexible ribbon cable. The Camera Serial Interface port is specifically designed for high-speed data transmission between the camera and the processor. The image sensor captures light and converts it into electrical signal which is digitized and processed by the Graphic Processor Unit. The processed image data is then transmitted from the camera module to the raspberry pi 4 over the CSI. The processed image is then displayed on the screen. Figure 8 below shows the raspberry pi 4 which is the selected microprocessor.



Figure 6: Raspberry pi 4

4. **Electronic speed controllers (ESC)**

Electronic speed controllers (ESC) regulate the speed and direction of motors based on the input signals. Upon signal reception from the raspberry pi 4, the microcontroller inside the ESC interprets the input signal and determines how much power to send to the motor thus varying the width of the Pulse Width Modulation (PWM). The Metal Oxide Semiconductor Field Effect Transistor (MOSFET) rapidly switches the DC power received by the ESC to the motor's 3 Phase AC signal in a sequence. They include a pair of relays for each motor. The sensors required for the inspection are added as payload and integrated with the circuitry. Ethernet cable is used as the tether for transmission of control signals, the camera feed, and the sensor data from the ROV. Various sensors specific to the application are equipped to carry out detailed underwater monitoring. These sensors also serve as a health monitor to prevent its navigation beyond the operating range.

The brushless motor has a rotor which is the permanent magnet and the stator which has copper coils. When the current is delivered to the coil of the stator, the coil becomes an electromagnet and develops polarities which repel the rotor magnet causing it spin. The 3 Phase AC current delivered by the ESC to the motor constantly changes the polarity of the electromagnet and this keeps the rotor spinning. Hall sensors incorporated with the motor precisely track the rotor's position. Figure 9 below depicts the electronic speed controller.

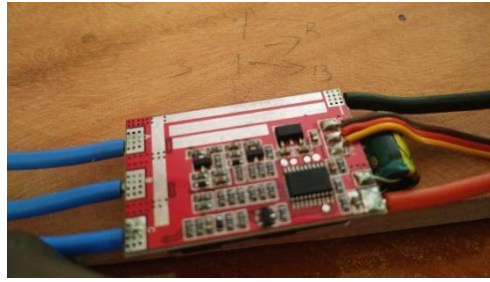


Figure 7: Electronic speed controller

5. Brushless motor

It has a rotor which is the permanent magnet and the stator which has copper coils. When the current is delivered to the coil of the stator, the coil becomes an electromagnet and develops polarities which repel the rotor magnet causing it spin. The 3 Phase AC current delivered by the ESC to the motor constantly changes the polarity of the electromagnet and this keeps the rotor spinning. Hall sensors incorporated with the motor precisely track the rotor's position.

The ROV is equipped with a temperature sensor with a usable temperature to watch the temperature underwater for any alarming temperature variations. The pressure sensor is used to provide real-time pressure data that helps in determining the depth of the ROV. It also serves as an external aiding sensor for position tracking by providing depth. The data from these sensors are serially communicated to the microcontroller which is connected to the processor. Figure 10 shows the brushless motor with the propeller attached to it.

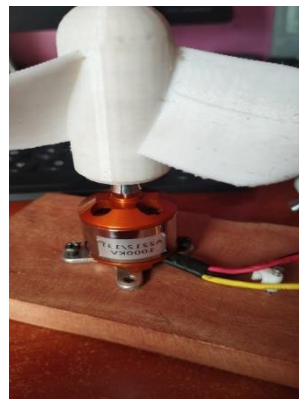


Figure 8: Brushless motor with propellor attached

6. The Inertial Measurement Unit (IMU)

It contains a gyroscope, accelerometer, and a magnetometer for providing compensated incremental angle, velocity data for navigation and angular rates, linear accelerations, and magnetic fields for control through a digital serial interface bus. The IMU measures angular rates(rad/sec), linear acceleration(m/sec^2), and magnetic fields(milli-gauss) in a body mounted strap down configuration

along the three axes which are fed to a sensor fusion algorithm to obtain the attitude and heading of the ROV with respect to the Earth Centred Inertial frame. The navigation data involving the incremental angles and velocities along the three axes are primarily used for positioning the ROV while the control data is used in the determination of the orientation in Euler angles (roll, pitch, and yaw) and control of the vehicle. Since stand-alone gyroscope has no frame of reference in contrast to accelerometer with earth's acceleration due to gravity as the reference and magnetometer with the earth's magnetic north as a frame of reference, all the three sensors are fused to obtain the complete orientation of the sensor with respect to a navigation frame. The IMU is powered with a 5V converter. It is mounted vertically on the flat tray at the centre of gravity of the vehicle such that the x-axis of the IMU coincides with the direction of the vehicle's heading. Further calibration of the IMU after placing it at the COG is done to reduce the bias and obtain precise values. Figure 11 below show Mpu6050- Inertial Measurement Unit.

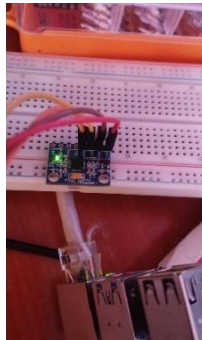


Figure 9: Mpu6050: Inertial Measurement Units

7. Tether management system

ROV's use tether, high-bandwidth communication cable for communication with the operator. The fibre optic tether enables the transfer of high-definition videos. The figure below depicts the ethernet cable used for data transmission. Figure 12 shows the ethernet cable which is used for transmission of data.



Figure 10: An ethernet cable used for transmission of data

3.2 Fabrication and assembly

Implementing the ROV Electrical and Control design followed a series of processes that involved acquisition of electrical components and integration of the software and hardware, and the assembly of the ROV.

3.2.1 Project Setup

1. Requirements

- Git
- Python version 3.10
- An Editor preferable PyCharm
- Laptop Specs: 4GB Ram, i5/i3/i7 processor

2. Local set up

- The Project Source Code is publicly located at

https://github.com/dennissammy77/REMOTELY_OPERATED_VEHICLE.git

- The following commands were performed:

```
git clone https://github.com/dennissammy77/REMOTELY_OPERATED_VEHICLE.git
```

- The required dependencies were installed as listed in the main.py file.

The file inventory of the Raspberry Pi was accessed and the server folder of the ROV was copied to the raspberry Pi under the path "~/home/<user>/ROV."

3. Secure Socket Shell (SSH) Configuration

Secure Socket Shell connection enable the Graphic User Interface to connect to the Raspberry Pi 4 thus allowing the microprocessor to operate and be accessed remotely and headless which refer to connection without an interface.

This setup involved the following steps:

- Downloading the Raspberry Pi 4 installation imager.
- Recording the IP, username and password during Raspberry pi OS installation. Username and password were used as authenticators for the SSH remote headless connection.

- Under the program repository, the respective values to the variables were assigned.

4. Run the Program

With the entry point of the program as main.py, the program was started with either of the following methods:

- Accessing the location of the program and entering the below command at the operator's terminal.

```
python3 main.py
```

- Running the file in the debugger mode.

5. Modules installation

The required modules were stated at the entry file, main.py. The following commands were run at the operator's terminal to install dependencies.

```
pip3 install <name_of_module>
```

Also, the terminal of the raspberry pi 4 was assessed and ensured that the modules were installed. An alternative in the event the above command did not apply is the command below:

```
sudo pip install <name_of_module>
```

or

```
sudo git clone <module_repo>
```

6. Electrical setup

The respective signal pins indicated in the services files were connected to the appropriate GPIO pinout pins on the raspberry pi. A breadboard was used to ensure to ensure a firm connection. The sensors and the electrical components were connected to the signal pins.

The precaution when connecting the ESC to the GPIO pins on the Raspberry Pi is not to connect the middle pin to the Raspberry Pi. This is to protect the microprocessor from over voltages.

7. Serial Connection

Socket protocol communication was enabled by tether/ethernet cable. The ethernet cable was ensured to be firmly connected.

8. Thruster connection

The ESC was connected to the motor. It was powered by a secondary power, LiPo batteries. The connections were clearly marked with the Red and Black wires. The signal connection needed a GPIO input from the Raspberry PI and a ground connection from the Pi. The opposite three connections on the ESC were connected to the brushless Motors. Reversing these connections consequently reversed the direction of the motor rotation.

3.3 Testing

Testing - a critical phase in the design of a ROV – ensured the functionality of the designed control and electrical system of the ROV to achieve manoeuvrability in the main axes of motion. It was conducted to validate the motor control's functionality, performance, and reliability. The setup included a Raspberry Pi 4 with GPIO pins configured for motor control, motor mount, a power supply, and a client device for sending commands. The testing commenced by validating the connection between the client and the motor control server, ensuring that it was ready to receive commands. The motor was tested for its ability to start and stop using basic commands, such as forward and reverse, while observing its responsiveness. Variable speed tests followed, where different speed percentages (e.g., 25%, 50%, 75%) were sent to verify the motor operates as expected. Edge case scenarios, such as sending invalid commands or rapid command changes, were also tested to ensure the system handled errors well while remaining stable. An emergency stop test confirmed the motor could halt immediately when required, demonstrating the system's safety features. The testing demonstrated smooth speed transitions, robust error handling, and consistent stopping behaviour. These provided insights into the system's stability, performance, and scalability, while identifying any areas for its improvement to ensure the system was ready for integration.

4. Results and Discussion

4.1 Research and selection of potential materials for key components.

The ROV is to conduct hull inspection on a vessel and thus the material selected should withstand the pressure experienced at the draught of the vessel. Settling on an average depth of 10m, which is greater than the draughts of majority small marine vessels such as Aylah 1 and Falcao, the pressure is calculated.

4.2 Analysis of proposed materials

The model development study was carried out on the existing ROV of the dimensions given in table 3 and it was modelled to the design plan. A smaller frame that still has minimal solid obstructions is ideal yet still places the thrusters in a manner to ensure manoeuvrability in the respective degree of freedom.

Table 3: Principal Dimension of the ROV

No.	Dimension	Overall Size (mm)
1.	Length	550
2.	Width	550
3.	Height	211.13
4.	Thickness	5

The design considerations for the frame are:

- Simplicity and rigidity
- Symmetry in thrusters' distribution to minimize undesired reaction moments.
- The vehicle should be nearly neutrally buoyant, and the centre of buoyancy should be placed above the centre of gravity to generate righting moment.

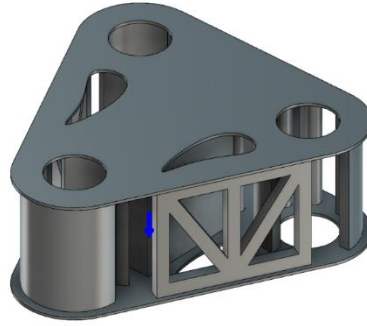


Figure 11: Frame model

The analysis was taken through a series of 3 depth variations i.e. 10m, 50m, and 100m. Table 4 below shows the hydrostatic load used.

Table 4: Pressures at different depths

No	Depth	Load (Pa)
1	10	101,043
2	50	505,215
3	100	1,010,430

The first analysis was carried out on each material to establish variation effect to the model with the load from 10m water depth and the results were tabulated in table 12, 13, 14 and 15 below. Figures 16 and 17 below provide stress and deformation comparisons for the materials at a depth of 10 m.

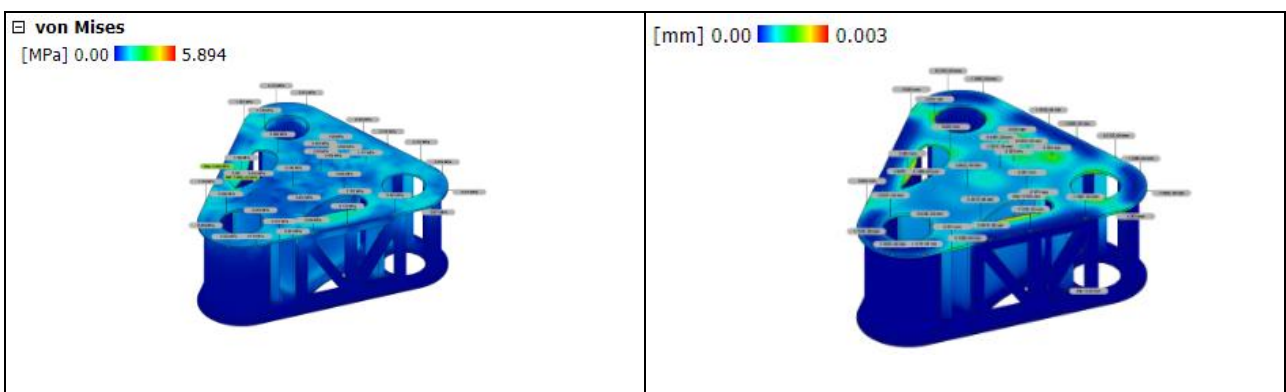


Figure 12: Von Mises Stress and Deformation Using Steel

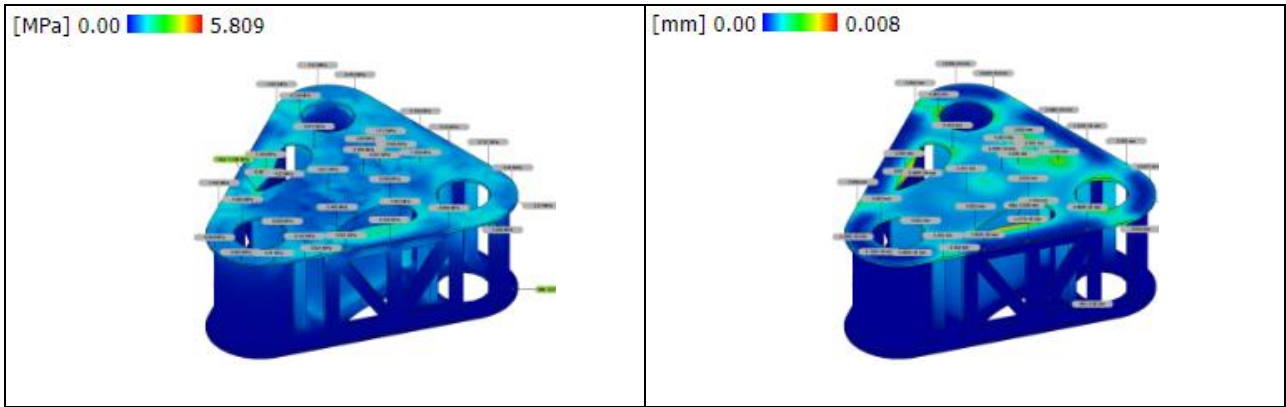


Figure 13: Von Mises Stress and Deformation Using aluminium

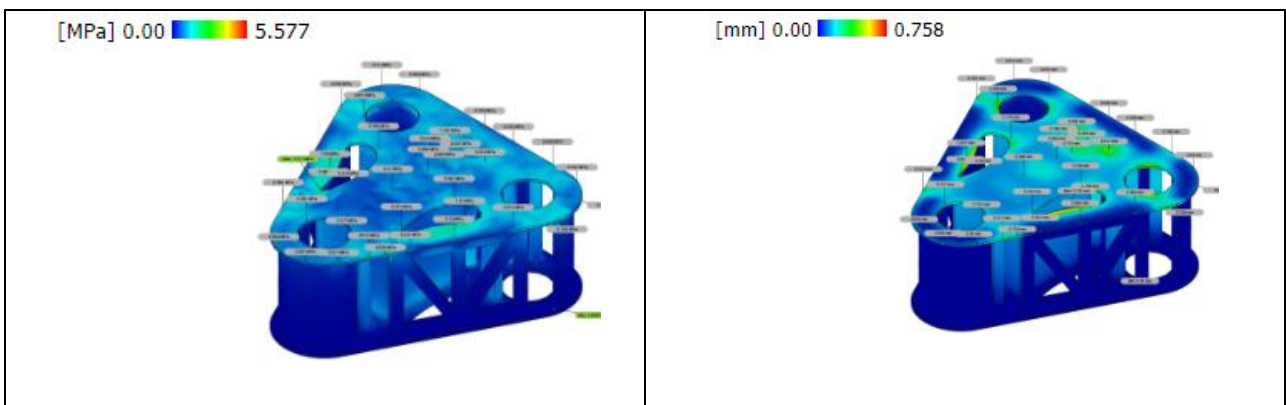


Figure 14: Von Mises Stress and Deformation using PVC.

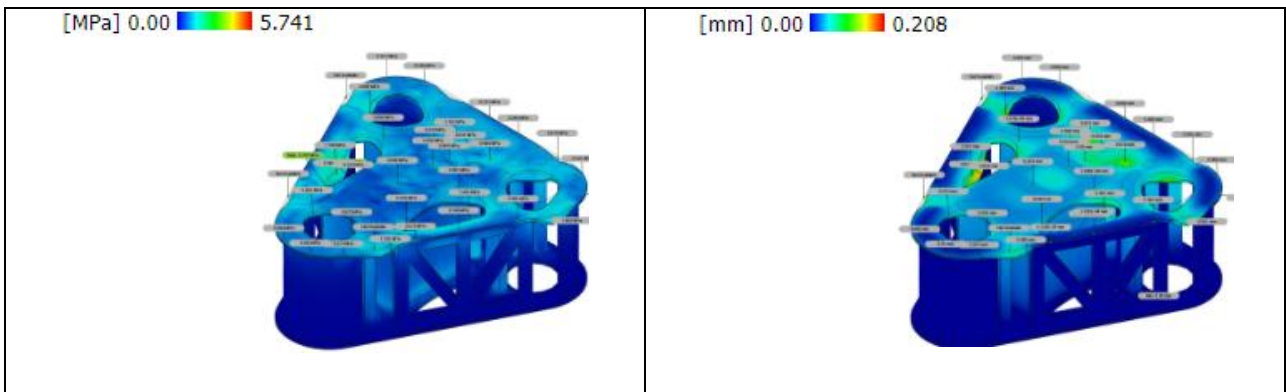


Figure 15: Von Mises Stress and Deformation Using Acrylic

Table 5: Von Mises Stress and Deformation of different materials at 10m

No	Material	Deformation (mm)	Deformation (sigmoid function)	Stress (MPa)	Stress (sigmoid function)
1	Steel	0.003	0.5007499994	5.894	0.9972516434
2	Aluminium	0.008	0.5019999893	5.809	0.9970085466

3	PVC	0.758	0.6809193547	5.577	0.9962303637
4	Acrylic	0.208	0.5518133302	5.741	0.996798726

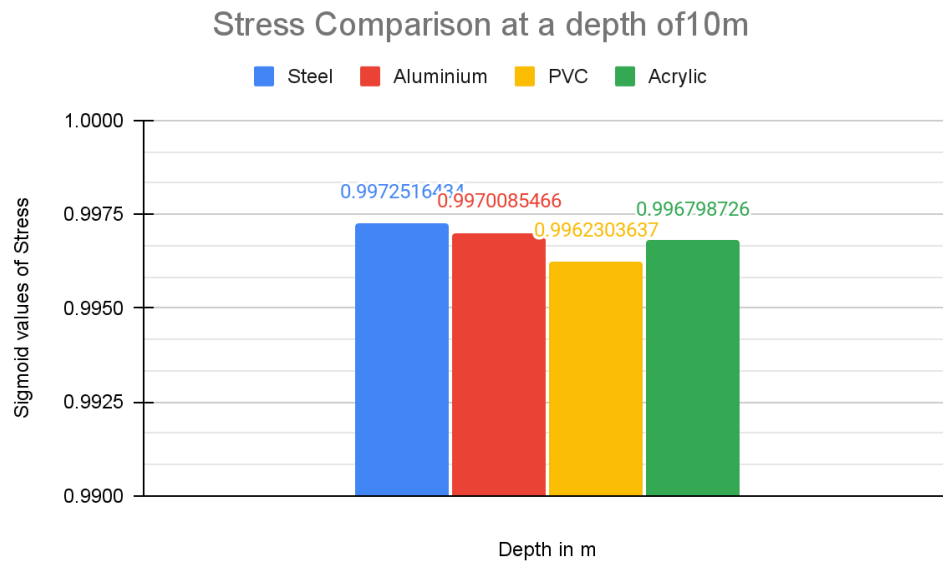


Figure 16: Stress comparison at a depth of 10 m

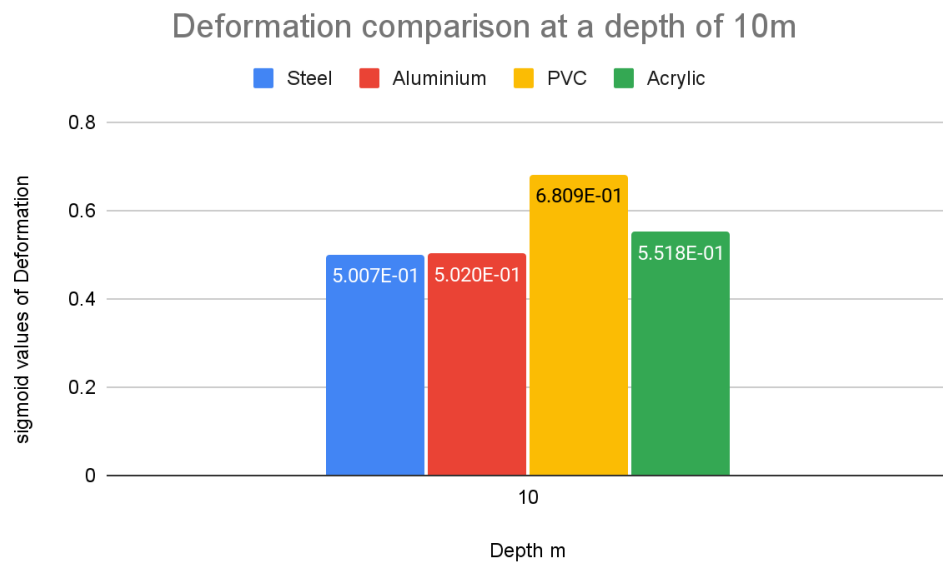


Figure 17: Deformation comparison at a depth of 10 m

The stress and deformation results of each material at depths of 10m, 50m and 100m are shown in tables 6, 7, 8, 9, 10, 11, 12, and 13 below. Table 14 below depicts the properties of each material

while table 15 below shows the allowable stresses of each material. Figure 18 and 19 below depicts deformation and stress comparison of the materials respectively at depths of 10m, 50, and 100m.

Table 6: Stress and Deformation of Steel on Depth Variation

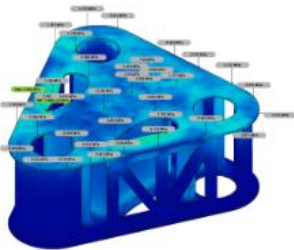
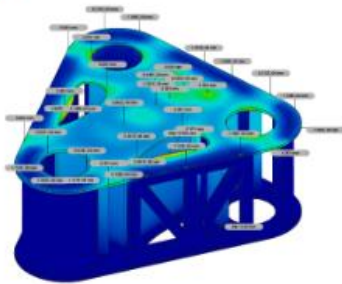
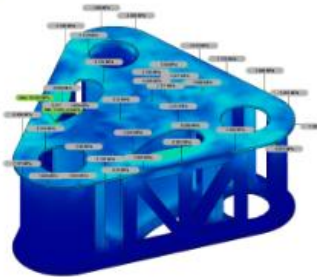
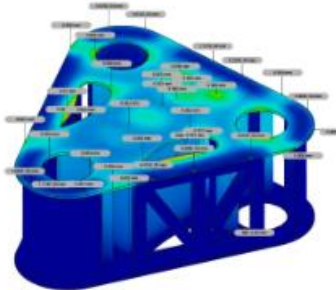
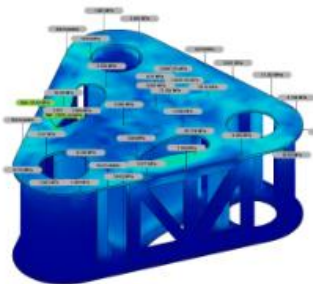
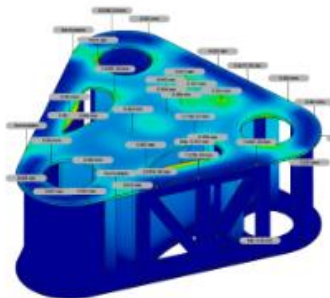
Depth (m)	Stress analysis	Deformation analysis
10	☐ von Mises [MPa] 0.00 5.894 	[mm] 0.00 0.003 
50	[MPa] 0.00 29.433 	[mm] 0.00 0.015 
100	[MPa] 0.00 58.83 	[mm] 0.00 0.031 

Table 7: Stress and Deformation of Steel on Depth Variation - values

STAINLESS STEEL						
No	Depth (m)	Load (Mpa)	Deformation (mm)	Deformation (sigmoid)	Stress (Mpa)	Stress (sigmoid)
1	10	0.101043	3.00E-03	5.007E-01	5.894	0.997252
2	50	0.505215	1.50E-02	5.037E-01	29.433	1.000000
3	100	1.01043	3.10E-02	5.077E-01	58.83	1.000000

Table 8: Stress and Deformation of Aluminium on Depth Variation

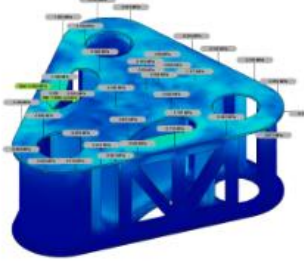
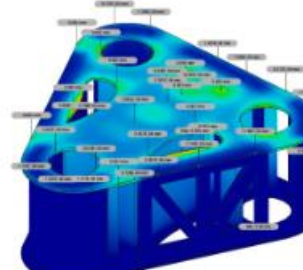
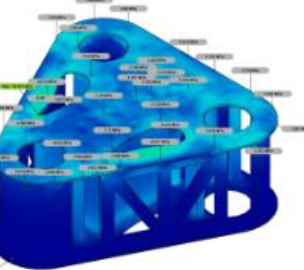
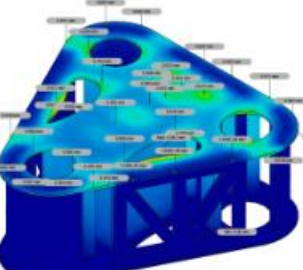
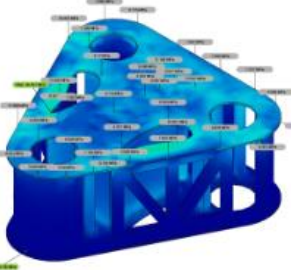
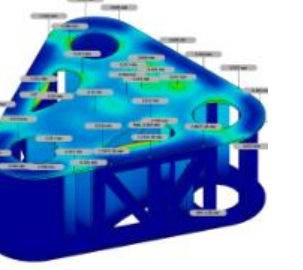
Depth (m)	Stress analysis	Deformation analysis
10	[MPa] 0.00  5.894 	[mm] 0.00  0.003 
50	[MPa] 0.00  29.042 	[mm] 0.00  0.042 
100	[MPa] 0.00  58.057 	[mm] 0.00  0.084 

Table 9: Stress and Deformation of aluminium on Depth Variation - values

ALUMINIUM						
No	Depth (m)	Load (Mpa)	Deformation (mm)	Deformation (sigmoid)	Stress (Mpa)	Stress (sigmoid)
1	10	0.101043	8.00E-03	5.020E-01	5.809	0.997009
2	50	0.505215	4.20E-02	5.105E-01	29.042	1.000000
3	100	1.01043	8.40E-02	5.210E-01	58.057	1.000000

Table 10: Stress and Deformation of PVC on depth Variation.

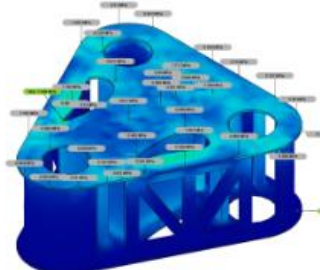
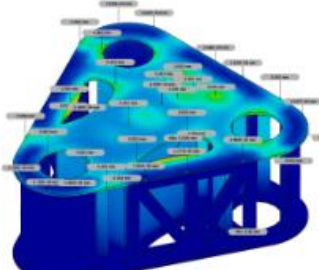
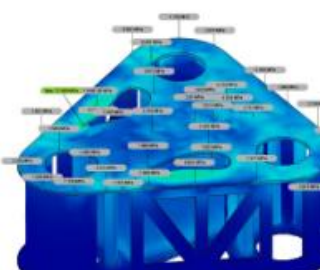
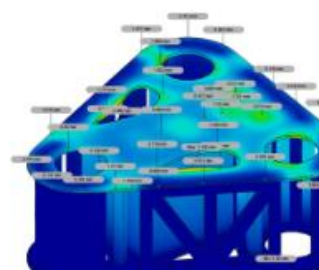
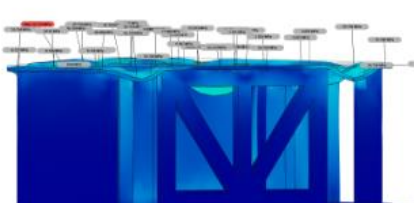
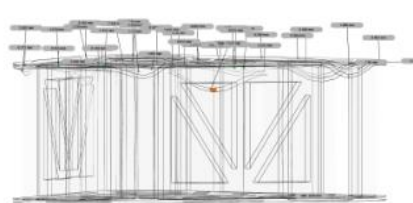
Depth (m)	Stress analysis	Deformation analysis
10	[MPa] 0.00  5.809 	[mm] 0.00  0.008 
50	[MPa] 0.00  27.889 	[mm] 0.00  3.788 
100	[MPa] 0.00  55.754 	[mm] 0.00  7.572, Threshold: 7.351 - 7.572 

Table 11: Stress and Deformation of PVC depth variation - values

PVC UNPLASTICIZED						
No	Depth (m)	Load (Mpa)	Deformation (mm)	Deformation (sigmoid)	Stress (Mpa)	Stress (sigmoid)
1	10	0.101042	0.758	6.809E-01	5.577	0.996230
2	50	0.505215	3.788	9.779E-01	27.889	1.000000
3	100	1.01043	7.572	9.995E-01	55.754	1.000000

Table 12: Stress and Deformation of Acrylic on depth Variation

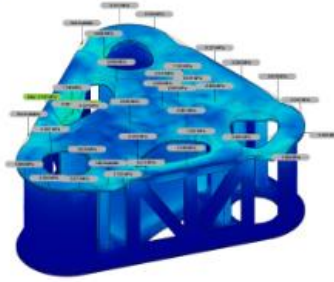
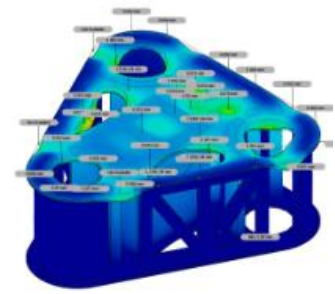
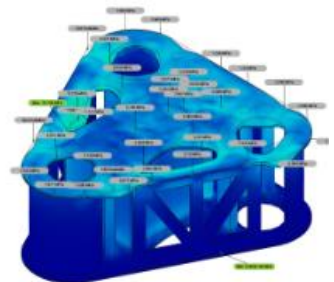
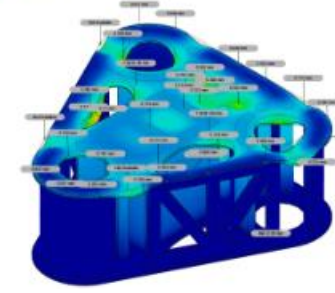
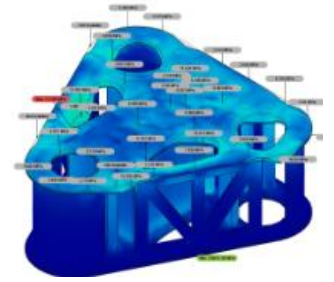
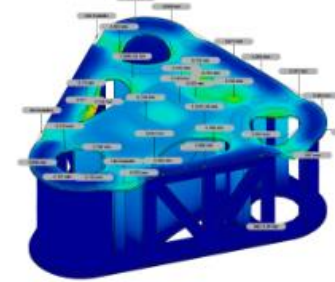
Depth (m)	Stress analysis	Deformation analysis
10	[MPa] 0.00  5.741 	[mm] 0.00  0.208 
50	[MPa] 0.00  28.709 	[mm] 0.00  1.042 
100	[MPa] 0.00  57.394 	[mm] 0.00  2.083 

Table 13: Stress and Deformation of Acrylic on depth variation - values

ACRYLIC						
No	Depth (m)	Load (Mpa)	Deformation (mm)	Deformation (sigmoid)	Stress (Mpa)	Stress (sigmoid)
1	10	0.101042	2.08E-01	5.518E-01	5.741	0.996799
2	50	0.505215	1.042	7.392E-01	28.709	1.000000
3	100	1.01043	2.083	8.892E-01	57.394	1.000000

Stress and deformation limitations are calculated to determine the value of stress and deformation permitted to the material installed. Structure stress and deformation that overvalue the allowable limit are not recommended to be used in the design.

Allowable stress can be expressed by:

$$\sigma = \frac{\text{Ultimate Tensile Strength}}{\text{Safety Factor}} \quad 10$$

Table 14: Properties of materials

Properties	Steel	Aluminium	PVC	Acrylic
Density	7.850E-06 kg / mm ³	2.700E-06 kg / mm ³	1.300E-06 kg / mm ³	1.188E-06 kg / mm ³
Young's Modulus	210000.00 MPa	68900.00 MPa	6.97 MPa	2740.00 MPa
Poisson's Ratio	0.30	0.33	0.39	0.355
Yield Strength	207.00 MPa	275.00 MPa	11.00 MPa	48.90 MPa
Ultimate Tensile Strength	345.00 MPa	310.00 MPa	11.00 MPa	79.80 MPa

Table 15: Allowable stress for the different materials

No	Material	Stress (MPa)
1	Steel	115
2	Aluminium	77.5
3	PVC	18.75
4	Acrylic	10.4



Figure 18: Deformation comparison of each material at different depths

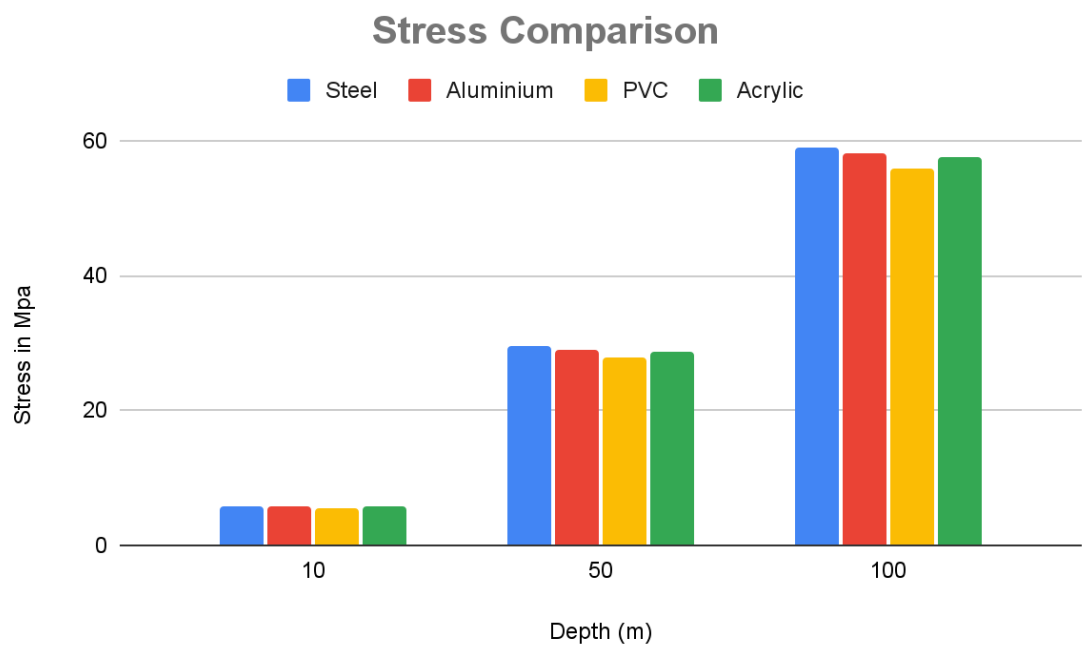


Figure 19: Stress comparison of each material at different depths

The analysis shows that the stress value of the ROV with every material variation is within allowable values in all operational conditions. The differences in stress and deformation are caused by the values of yield strength and modulus of elasticity that affect the material. However, aluminium is susceptible to corrosion in marine environment and require treatments such as anodizing which add

to the cost and maintenance requirements. Further, aluminium requires techniques for welding and precise machining. Acrylic offers superior UV resistance than PVC preventing degradation from sunlight exposure. It is also lightweight in comparison to PVC and has a polished appearance.

The Sealing Compartment and End Caps

The acrylic tube is a cylinder of 400mm length and an exterior diameter of 110mm. The acrylic tube's inner diameter is about 105mm as is seen in the construction of Miniature ROVs [30].

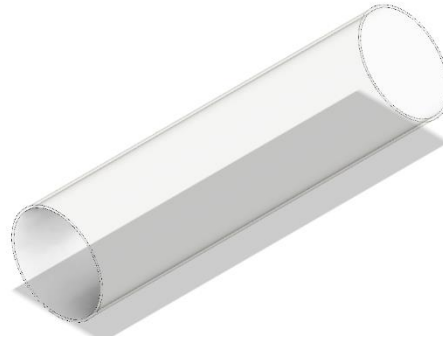


Figure 20: Modelled acrylic sealing compartment

The sealing compartment is subjected to a stress and deformation analysis and the results are tabulated in figure 21 and table 16 below. Figure 22 shows the designed end cap.

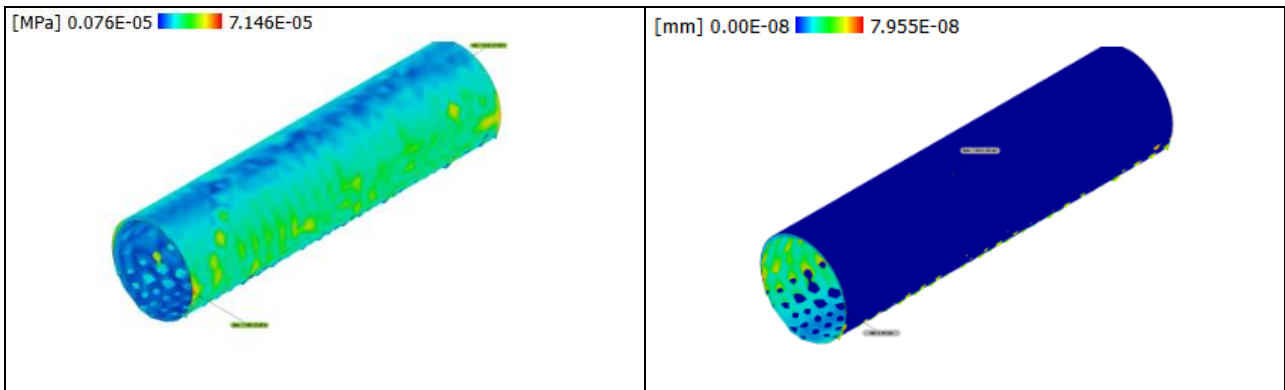


Figure 21: Stress and Deformation of Acrylic sealing compartment at 100m depth.

Table 16: Stress and deformation of acrylic on depth variation - values

No	Depth(m)	Load (Mpa)	Deformation(mm)	Stress (Mpa)
1	100	1.010430	7.955E-08	7.146E-05

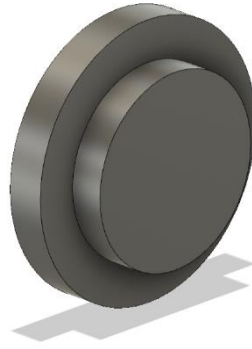


Figure 22: End Caps

It is observed that the acrylic cylindrical sealing compartment holds under the pressure experienced at a depth of 100m.

4.3 Determining the weight and displacement of the ROV.

The information on the weight of the ROV is vital for designing the buoyancy system and ensuring that the ROV operates effectively under water. Table 17 below shows the ROV components, and their corresponding mass obtained upon research of the components not acquired during the current work and by use of weighing scale for the already acquired electrical components. It also depicts the total weight of the ROV.

Table 17: Total Mass of the ROV

Item	Qty	Mass [Kg]	Size (mm)	Total mass [Kg]
Frame				
Acrylic Plexiglass	1	0.45	400 x D110	0.45
O-ring seals	6	0.086	-	0.516
Frame	1	13.39325	550 x 550 x 210.11	13.39325
Electrical				
Battery for motors	6	0.2	94 x 94	1.2
Battery for electrical components	1	0.2		0.2
Raspberry Pi 4	1	0.1	8.5× 5.6× 2	0.1

Heatsinks for raspberry Pi 4	3	0.05	14 x 14 x 6	0.15
Relays for motors	6	0.015	70 × 50 × 20	0.09
ESC 30A	6	0.03	139 x 28	0.18
Brushless Motor	6	0.06	50 x 30 x 30	0.36
Camera	1	0.063	70 x 50.8 x 43.9	0.063
LED lights	2	0.05		0.1
Led driver	2	0.004	25.3 x 25.3 x 10.3	0.08
waterproof LED Lens	2	0.025		0.05
SD card	1	0.002	32.0 × 24.0 × 2.1	0.002
USB-type A connector	1	0.04		0.04
MOSFET IRF1404	1	0.02	5 x 2.76 x 0.31	0.02
Pressure sensor	1	0.02		0.02
Nylon plastic	1	0.5		0.5
Pixhawk controller	1	0.05		0.05
Motor driver	2	0.06	56 x 57	0.12
Pulse width modulation	6	0.09	100 × 80 × 40	0.54
Propulsion				
Thrusters	6	0.075	72 x 92	0.45
		Total Mass		18.67425
		Total weight		183.1943925

From table 22, the estimated mass of the ROV is 18.67425kg, estimated weight, taking gravitational pull as 9.81m/s^2 , is 183.1943925N, and the volume of displacement is $1.127\text{E}+7\text{mm}^3$.

4.4 Calculating the buoyant force, hydrodynamic drag force, power of the motor, and the thrust Force.

The factors considered in the selection of a suitable motor for the ROV to achieve a designed speed of 2m/s include the hydrodynamic drag, thrust required and power needed.

4.4.1 ROV Upthrust Force

To find the force of upthrust experienced by the ROV, Equation (11) is used.

$$18.76425 \times 9.81 = 183.1493925 \text{ N} \quad 11$$

4.4.2 ROV buoyancy

To find the force of buoyancy experienced by the ROV, Equation (12) is used

$$F_b = 8.0\text{E}-6 \text{ kg/m}^3 \times 9.81 \times 0.01127 \quad 12$$

$$F_b = 8.84\text{e}-7 \text{ N} \quad 13$$

4.4.3 ROV hydrodynamic drag

The thrust required to achieve a speed of 0.5m/s depends on the hydrodynamic drag force acting on the ROV. The drag force on an ROV moving through water can be calculated using the drag Equation (17) below.

The Reynolds number is used to predict the flow patterns in fluid dynamics. It is a relation between density ρ , characteristic length L , velocity, U and fluid viscosity μ . The Reynolds number for the ROV in the water with $\rho = 1025 \text{ kg/m}^3$ and $\mu = 1.31 \text{ Cp} = 1.31 \times 10^{-3} \text{ kg/ms}$ and moving at 0.5 m/s is calculated below.

$$R_e = \frac{\rho UL}{\mu} \quad 14$$

$$R_e = \frac{1025 \times 0.5 \times 0.55}{0.00131} \quad 15$$

$$R_e = 215171.7557 \quad 16$$

The Reynolds number exceed 4000 signifying the flow to be turbulent. For turbulent flow around a smooth sphere, the coefficient of drag levels off to an approximately constant value as the Reynolds number increases, particularly beyond $Re \approx 10^5$.

The force of drag on the ROV vehicle is thus calculated by taking the values below:

Designed ROV speed = 0.5m/s

Density of sea water = 1025kg/m³

Wetted surface area of the ROV = 1.763m²

Coefficient of drag = 0.03

$$F_{D(Body)} = \frac{1}{2} \times 1025 \times 0.03 \times 1.763 \times 0.5^2 \quad 17$$

$$F_{D(Body)} = 6.776N \quad 18$$

The total drag of the ROV system is the vehicle drag, and the tether drag. The coefficient of drag for tether cables used in Remotely Operated Vehicle (ROV) applications vary depending on the cable's surface characteristics, diameter, and flow conditions. The tether drag coefficient in a turbulent flow regime is approximated to be 1.2 [31]. With a tether cable of 0.01m and a length of 10m, the characteristic area of the tether cable is obtained as 0.004712388m². The drag force on the tether cable is thus calculated in the Equation (19).

$$F_{D(Tether)} = \frac{1}{2} \times 1025 \times 1.2 \times 0.004712388 \times 0.5^2 \quad 19$$

$$F_{D(Tether)} = 7.245N \quad 20$$

The total hydrodynamic drag force on the ROV system is thus the summation of the tether drag, and the vehicle drag, and it is given by Equation (21) below.

$$F_{D(Total)} = 7.245 + 6.776 = 14.02N \quad 21$$

Power analysis of motor

To calculate the power of a motor needed to achieve a maximum speed of 1m/s and overcome a drag force of 14.02N, the fundamental relationship between power, force and velocity is applied in Equation (22).

$$P_{(req)} = 14.02 \times 1 \quad 22$$

$$P_{(req)} = 14.02 \quad 23$$

By including 20% losses, the maximum power of the motor is obtained from equation:

$$P_{\max} = 14.02 \times 1.2 \quad 24$$

$$P_{\max} = 16.824 \text{ W} \quad 25$$

The motors need to produce 16.824W, which is equivalent to 0.022555992hp to maintain a maximum speed of 1m/s while overcoming the ROV system drag force of 14.02N. It is assumed that the drag force remains constant at the designed speed.

4.5 The control system functional implementation

The developed ROV control system consists of a user-friendly Graphical User Interface (GUI), a robust system code, and an integrated communication protocol. The system allows seamless operation of the vehicle, enabling users to control its movements (e.g., forward, backward, turning) in real-time. The integration of the GUI with the system code ensures smooth and efficient operation with minimal latency.

4.5.1 Key features demonstrated

- The GUI provided intuitive controls for user commands.
- System code effectively processed commands and transmitted them to the system onboard the vehicle.
- Real-time control and feedback mechanisms ensured reliable operation.

4.5.2 Performance metrics

The system was evaluated based on several performance indicators, as outlined in table 18 below:

Table 18: Performance metrics of control system

Metric	Measured Value	Target Value	Remarks
Response Time	2s	< 1s	System does not meet expected response time.
Command Accuracy	50%	> 90%	Medium accuracy achieved in movement commands.
Communication Latency	2s	$\leq$ 250 ms	No instant communication
Stability	20 min	$\geq$ 1hr	System runs without errors or crashes.

4.5.3 System demonstration and Results

1. Basic movement commands:

- **Top Command:** The Motor representing the top direction movement started rotation after the commands were made with a response time of 2s.
- **Turning Command:** The Motor representing the Left direction movement started rotation after the commands were made with a response time of 3ms.
- **Stop Command:** Immediate halting with a response time of 30s.

2. Stress testing:

- The system was tested for prolonged operation 20 minutes under varying command frequencies, maintaining consistent performance.

3. GUI and system code integration:

- Screenshots of the GUI in action are shown in figure 23.
- Logs from the communication system indicated smooth data transmission, with no dropped commands during testing.

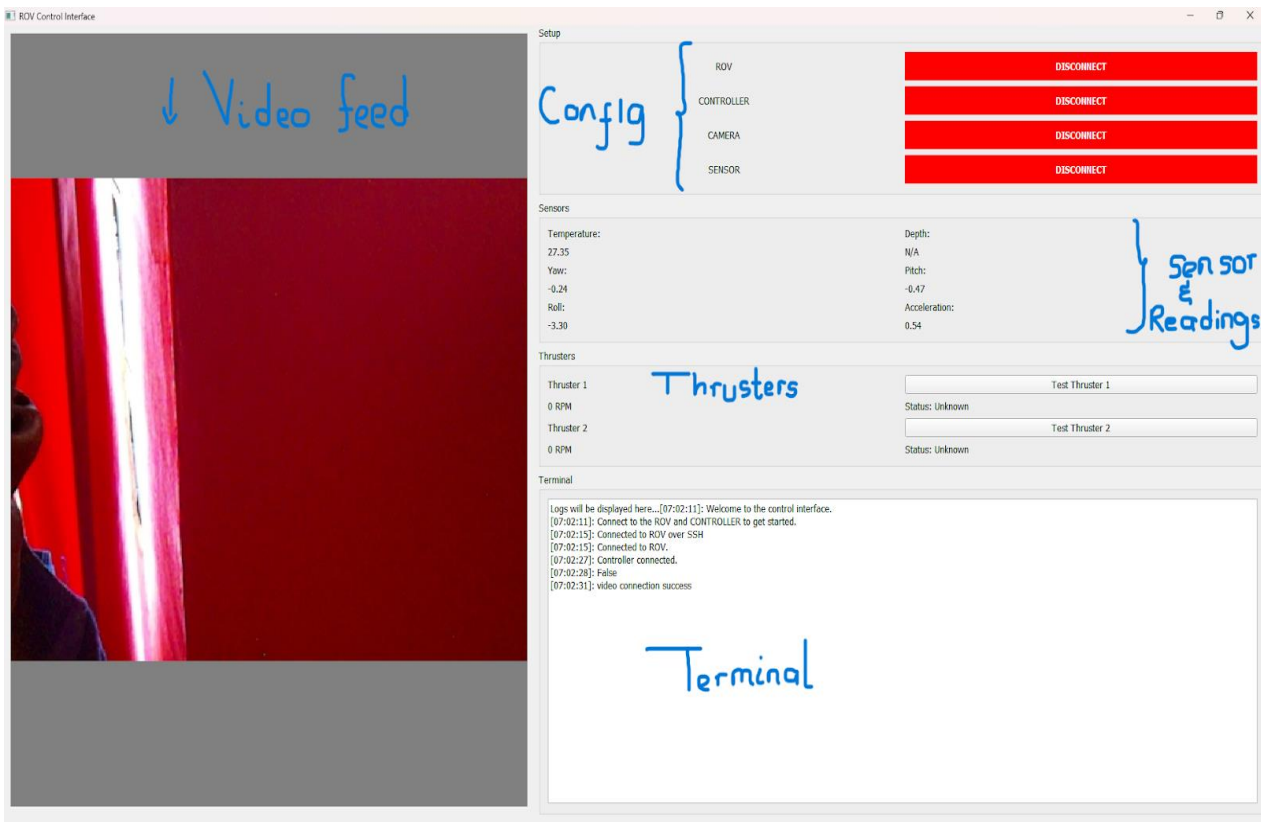


Figure 23: Screenshot of the GUI showing controls for movement and status indicators.

Figure 24 below depicts how the different components were assembled.

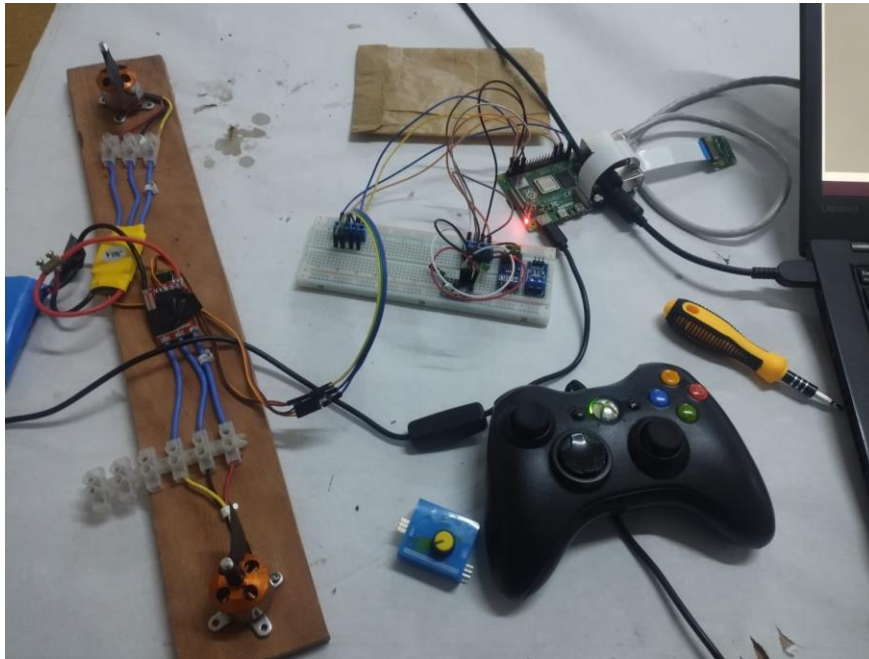


Figure 24: Assembly of the components

4.5.4 System performance analysis

The control system performed exceptionally average in terms of **response time**, **command accuracy**, and **stability**:

- Response time was consistently above the target threshold of 1 second, with an average of **2s**.
- Command accuracy was measured at **50%**, indicating unstable and latency discrepancies in operation of the vehicle.

The GUI proved to be intuitive and effective, allowing users to control the ROV without requiring prior technical expertise.

4.6 Propeller

A three-bladed propeller was chosen to ensure smooth and stable propulsion. The tapered leading edge and the flat trailing edge are designed to minimize drag while maximizing thrust. High strength PLA material was used to 3D print the propellers due to its durability and resistance to deformation under water.

5 Conclusion

The design and analysis of the ROV proved a comprehensive approach to material selection, structural integrity, and control system implementation, ensuring its suitability for hull inspections in underwater environments. The electrical and control systems were critical components of the design, enabling the ROV to achieve functional precision and operational efficiency. The analysis of materials and their deformation under pressure showed that all selected materials remained within their allowable stress limits. Acrylic underwent a maximum stress deflection of 57.394 Mpa and a maximum deformation of 2.083mm at the depth of 100m. It's frame further provided a lightweight yet robust structure, while the electrical and control systems ensured functional accuracy, stability, and ease of operation. Additionally, the modular design of the ROV allows for future enhancements, such as improved sensors, additional payloads, or upgraded control algorithms. The control system was implemented to manage the ROV's thrusters for seamless manoeuvrability across six degrees of freedom. This system incorporates precise input handling from the controller, with commands processed and transmitted to the motor control circuits via ESCs (Electronic Speed Controllers). The use of a Raspberry Pi 4 as the central control unit facilitated seamless integration of sensor data, motor commands, and feedback loops, ensuring real-time adjustments for stability and performance.

Several challenges were encountered during the development of the project. They include:

1. The tests revealed a response time exceeding 2 seconds due to inefficient communication protocols.
2. During the first integration, mismatched data formats caused command failures. Debugging and standardizing data formats resolved the issue.
3. The selection of a unidirectional electronic speed controller constrained the achievability of the direction of motor to only one direction.

It is recommended to use a bidirectional Electronic Speed Controller.

6.Budget

Table 19: Budget

ITEMS	NUMBER OF ITEMS	COST PER ITEM (KSH)	TOTAL COST (KSH)
Brushless motors	2	1500	3000
ESC	2	1500	3000
propellor	2	600	1,200
Battery Li Po	2	1800	3600
Raspberry Pi 4	1	13700	13700
Camera Module	1	5200	5200
Jumper Cables	-	240	240
Connectors	4	60	240
Breadboard	1	400	400
IMU: MPU6050	1	600	600
Total			31180

7. Time plan

Table 20: Time Plan

	May	Jun	Jul	Aug	Sept	Oct	Nov	Dec	Jan
Report Documentation									
Main structural design									
Electronic design									
Raspberry implementation									
GUI development									
Integration & Testing									
Final Presentation									

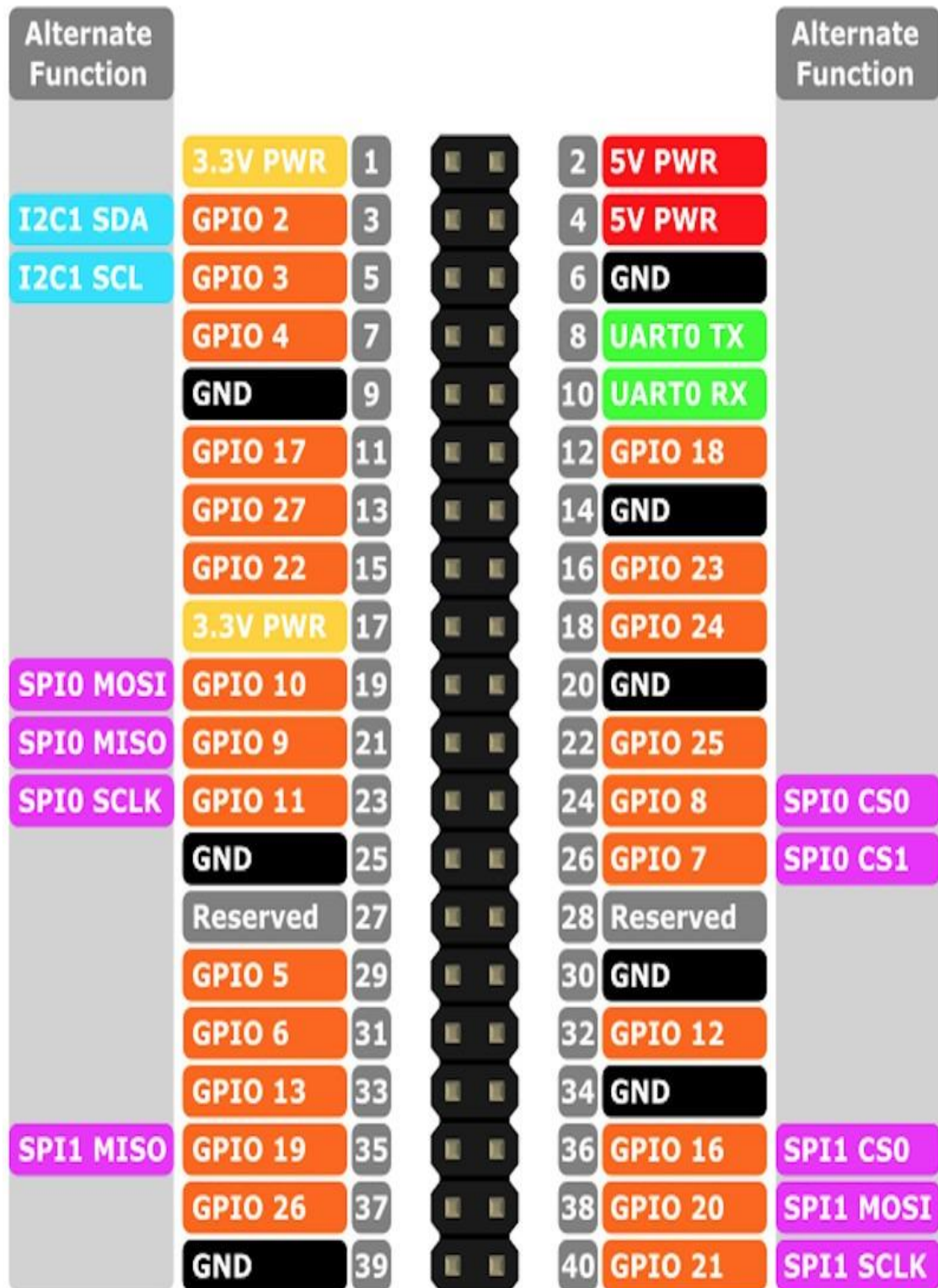
References

- [1] M. a. C. Agency, "In-water Surveys," Marine Guidance Note - MGN 546(M)."
- [2] F. L. Ivan Nagelkerken, et al., "Biofouling and its Impact on Maritime Fuel Efficiency and Emissions."
- [3] B. L. a. X. Dong, "Ship hull inspection: A survey Ocean Engineering," pp. vol. 289, p. 116281, 2023/12/01/ 2023, doi.
- [4] B. L. a. X. Dong, "Ship Hull Inspection: A Survey on Ocean Engineering," 11/05 2023.
- [5] N. L. a. S. Gautam, "Applications of Remotely Operated Vehicles (ROVs) in Offshore Operations: A Review on Recent Developments," 09/15 2023.
- [6] H. PARSONS, An insight into the history of ROVs.
- [7] B. Jacques, "Hempel launches underwater hull inspections using ROVs," S. Maritime, Ed., ed," Sept 2020.
- [8] a. B. S. O. Y. Y. Ahmed, " Jurnal Teknologi, vol. 69 :Design of a new low cost ROV vehicle," 07/20 2014.
- [9] P. I. M. e. al., "Eyes in the sea: unlocking the mysteries of the ocean using industrial, remotely operated vehicles (ROVs)," Science of the Total Environment," vol. vol. 634, pp. pp. 1077-1091., 2018.
- [10] NOAA, "Remotely Operated Vehicle," Ocean Engineering."
- [11] A. Wylie, "ROV Ventana completes 4500 Dives Deep Sea Dives."
- [12] a. D. W. D. J. I. Hudson, "A review of the uses of work-class ROVs for the benefits of science: lessons learned from the SERPENT project Underwater Technology," vol. vol. 26, no. 3, pp. pp. 83-88, . 2005.
- [13] a. J. W. G. P. J. Lever, "Benefits of Using Remotely Operated Vehicles to Inspect USACE Navigation Structures," p. p. 28, 03/01 2007.
- [14] E. Allmendinger, "Submersible vehicle systems de- sign. The society of naval architects and marine engineers, Jersey City (NJ)," 1990.
- [15] J. Smith, "Principles of Marine Engineering. New York: Maritime Press," 2010.
- [16] A. O. M. D. P. Zajac, "Stability Testing Method for Underwater ROVs," no. Polish Hyperbaric Research vol. 1(74)2021, 2021.
- [17] T. Wilson, "Underwater Robotics: Principles and Practices. Cambridge: Marine Tech Publishers," 2018.

- [18] M. Jones, "Advances in Underwater Robotics. San Francisco: Oceanic Research Publications," 2017.
- [19] R. Davis, "Materials in Marine Engineering. London: Oceanic Materials Publishing.," 2016.
- [20] S. Miller, "Innovations in Underwater Vehicle Design. New York: Marine Engineering Press," 2019.
- [21] P. Anderson, "Materials for Marine Applications. Boston: Coastal Engineering Publications," 2020.
- [22] J. Taylor, "Dynamics of Underwater Vehicles. Houston: Marine Systems Press," 2021.
- [23] A. Walker, "Hydrodynamics and Marine Propulsion. San Diego: Oceanic Engineering Publications," 2018.
- [24] D. Harris, "Marine Propulsion Systems and Design. Seattle: Nautical Engineering Press. 2019.
- [25] L. Martin, "Underwater Vehicle Propulsion and Dynamics. Miami: Ocean Engineering Publications," 2020.
- [26] F. b. Fakurruddin, "DESIGN OF PROPULSION SYSTEM FOR UNDERWATER REMOTELY OPERATED VEHICLE (ROV)," p. p 4, 2009.
- [27] T. Nguyen, "Advanced Marine Propulsion and Hydrodynamics. Cambridge," 2021.
- [28] P. Clark, "Hydrodynamics of Autonomous Underwater Vehicles," 2022.
- [29] R. Johnson, "Fluid Dynamics of Underwater Tethers. Oxford," 2019.
- [30] B. J. Smith, T., & Wilson, L, "Design and Construction of Miniature ROVs. Journal of Marine Engineering and Technology," p 22(3), 45-58, 2018.
- [31] B. R. Munson, Young, D. F., Okiishi, T. H., & Huebsch, W. W., "Fundamentals of Fluid Mechanics (7th ed.). Hoboken, NJ: Wiley.," (2013).
- [32] A. Stevens, "Hydrodynamics of Underwater Tether Systems," (2017).

Appendix

1. GPIO Pinout raspberry pi 4



2. Thruster Service code

```
# !/usr/bin/env python3
# Modules
import RPi.GPIO as GPIO
from time import sleep, time
import socket
import threading
# sudo lsof -i :8486
# sudo kill pid
class ESCController:
    def __init__(self, pin, motor_name, min_pulse=1000, max_pulse=2000, frequency=50):
        self.min_pulse = min_pulse
        self.max_pulse = max_pulse
        self.frequency = frequency
        self.signal_pin = pin
        self.motor_name = motor_name

        GPIO.setwarnings(False)
        GPIO.setmode(GPIO.BCM)

        # Setup PWM pin for speed control
        GPIO.setup(pin, GPIO.OUT)
        self.pwm = GPIO.PWM(pin, frequency)
        self.pwm.start(0)

        # Calibrate ESC
        #self.calibrate()

    def calibrate(self):
        print(f"Calibrating {self.motor_name} ESC...")
        duty_cycle = self.pulse_to_duty(self.min_pulse)

        #self.pwm.ChangeDutyCycle(5)
        self.pwm.ChangeDutyCycle(duty_cycle)
        duty_cycle = self.pulse_to_duty(self.max_pulse)
        self.pwm.ChangeDutyCycle(duty_cycle)
        self.pwm.ChangeDutyCycle(0)
        print(f"{self.motor_name} ESC calibrated")

    def pulse_to_duty(self, pulse_width):
        """Convert pulse width (microseconds) to duty cycle percentage"""
        period = 1000000 / self.frequency
        return (pulse_width / period) * 100

    def set_speed(self, speed_percent):
        """Set ESC speed as percentage (-100 to 100)"""
        # Ensure speed is between -100 and 100
        speed_percent = max(-100, min(100, speed_percent))
        abs_speed = abs(speed_percent)

        # Calculate pulse width
        pulse_range = self.max_pulse - self.min_pulse
        neutral_pulse = (self.max_pulse + self.min_pulse) / 2

        # Calculate pulse width
        print('pulse_range', pulse_range)
        print('speed_percent', speed_percent)

        if speed_percent >= 0:
            # Forward speed (positive values)
            pulse_width = self.min_pulse + (pulse_range * speed_percent / 100)
            print('pulse_width_FR', pulse_width)
        else:
            # Reverse speed (negative values)
```

```

        #pulse_width = self.min_pulse - (pulse_range * abs_speed / 100)
        pulse_width = 0
        print('pulse_width_RV', pulse_width)

    duty_cycle = self.pulse_to_duty(pulse_width)

    print(f"Setting {self.motor_name} speed to {speed_percent}% ({duty_cycle}% duty cycle)")
    self.pwm.ChangeDutyCycle(duty_cycle)

class MotorController:
    def __init__(self):
        # Define motor configurations with GPIO pins
        motor_configs = {
            'top': {'pin': 12, 'name': 'Top Motor'},
            'rear': {'pin': 16, 'name': 'Rear Motor'},
            'left': {'pin': 11, 'name': 'Left Motor'},
            'right': {'pin': 13, 'name': 'Right Motor'}
        }

        # Initialize all ESC controllers
        self.motors = {}
        for motor_id, config in motor_configs.items():
            self.motors[motor_id] = ESCController(
                pin=config['pin'],
                motor_name=config['name']
            )

    def process_command(self, command):
        """Process commands for all motors"""
        try:
            # Parse command string: "motor_id:speed:value"
            parts = command.split(':')

            # Handle emergency stop
            if command == 'STOP_ALL':
                self.stop_all()
                return

            if len(parts) != 3:
                print(f"Invalid command format: {command}")
                return

            motor_id, action, value = parts

            if motor_id not in self.motors:
                print(f"Unknown motor ID: {motor_id}")
                return

            motor = self.motors[motor_id]

            if action == 'speed':
                try:
                    speed = int(value)
                    motor.set_speed(speed)
                except ValueError:
                    print(f"Invalid speed value: {value}")
            else:
                print(f"Unknown action: {action}")

        except Exception as e:
            print(f"Error processing command: {e}")

    def stop_all(self):
        """Emergency stop all motors"""
        print("Emergency stop - stopping all motors")

```

```

        for motor in self.motors.values():
            motor.set_speed(0)

def handle_client(client_socket, controller):
    try:
        print("Client connected, waiting for commands...")
        while True:
            command = client_socket.recv(1024).decode().strip()
            if not command:
                continue

            print(f"Received command: {command}")
            controller.process_command(command)

    except Exception as e:
        print(f"Error handling client: {e}")
    finally:
        controller.stop_all()
        client_socket.close()
        print("Client disconnected")

def thruster_setup():
    # Initialize the motor controller
    controller = MotorController()

    # Setup socket server
    server_socket = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
    server_socket.setsockopt(socket.SOL_SOCKET, socket.SO_REUSEADDR, 1)
    server_socket.bind(('0.0.0.0', 8486))
    server_socket.listen(5)
    print("Multi-motor thruster control server listening")

    try:
        while True:
            client, addr = server_socket.accept()
            print(f"Connection from {addr}")
            client_handler = threading.Thread(
                target=handle_client,
                args=(client, controller)
            )
            client_handler.start()
    except KeyboardInterrupt:
        print("\nShutting down server...")
    except Exception as e:
        print(f"Error Occured: {e}")
    finally:
        controller.stop_all()
        server_socket.close()
        GPIO.cleanup()

# Command line execution
if __name__ == '__main__':
    thruster_setup()

```

3. Video service code

```
from flask import Flask, Response, request
from picamera2 import Picamera2
import cv2
import json, os, signal

app = Flask(__name__)

# Global variables for control
camera = None
stream_active = False

def initialize_camera():
    global camera
    if camera is None:
        camera = Picamera2()
        camera.configure(camera.create_preview_configuration(main={"format": 'XRGB8888', "size": (640,
480)}))
        camera.start()

def generate_frames():
    global stream_active
    while stream_active:
        frame = camera.capture_array()
        ret, buffer = cv2.imencode('.jpg', frame)
        frame = buffer.tobytes()
        yield (b'--frame\r\n'
            b'Content-Type: image/jpeg\r\n\r\n' + frame + b'\r\n')

@app.route('/video_feed')
def video_feed():
    global stream_active
    if not stream_active:
        return "Stream is not active"
    return Response(generate_frames(), mimetype='multipart/x-mixed-replace; boundary=frame')

@app.route('/start_stream', methods=['GET'])
def start_stream():
    global stream_active, camera
    if camera is None:
        initialize_camera()
    stream_active = True
    return "Stream started"

@app.route('/stop_stream', methods=['GET'])
def stop_stream():
    global stream_active
    stream_active = False
    return "Stream stopped"

if __name__ == '__main__':
    initialize_camera()
    stream_active = True # Start with stream active
    app.run(host='0.0.0.0', port=5001)
```

4. Sensor Service code

```
from mpu6050 import mpu6050
import time
import socket
import threading
import json
import math
import sys

# Create a new Mpu6050 object
mpu6050 = mpu6050(0x68)

# Define a function to read the sensor data
def read_sensor_data():
    # Read the accelerometer values
    accelerometer_data = mpu6050.get_accel_data()

    # Read the gyroscope values
    gyroscope_data = mpu6050.get_gyro_data()

    # Read temp
    temperature = mpu6050.get_temp()

    return accelerometer_data, gyroscope_data, temperature

def get_y_rotation(x, y, z):
    radians = math.atan2(x, math.sqrt(y * y + z * z))
    return -math.degrees(radians)

def get_x_rotation(x, y, z):
    radians = math.atan2(y, math.sqrt(x * x + z * z))
    return math.degrees(radians)

def handle_client(client_socket):
    try:
        while True:
            # Read the sensor data

            accelerometer_data, gyroscope_data, temperature = read_sensor_data()

            # Calculate pitch and roll
            pitch = get_x_rotation(accelerometer_data['x'], accelerometer_data['y'],
accelerometer_data['z'])
            roll = get_y_rotation(accelerometer_data['x'], accelerometer_data['y'],
accelerometer_data['z'])
            # Yaw calculation (integrate gyro_z over time)

            print(f"Pitch: {pitch:.2f}, Roll: {roll:.2f}, Yaw: ")

            sensor_values = {
                'pitch': pitch,
                'roll': roll,
                'yaw': gyroscope_data['y'],
                'accel-x': accelerometer_data['x'],
                'temp': temperature
            }

            # Convert the dictionary to a JSON string and then encode it as bytes
            sensor_values_json = json.dumps(sensor_values).encode('utf-8')
            print(sensor_values_json)

            client_socket.sendall(sensor_values_json)
            time.sleep(1) # Send data every second (adjust as needed)
    except Exception as e:
```

```

        print(f"Error handling client: {e}")
    finally:
        client_socket.close()

def sensors_setup():
    server_socket = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
    server_socket.setsockopt(socket.SOL_SOCKET, socket.SO_REUSEADDR, 1)
    server_socket.bind(('0.0.0.0', 8487))
    server_socket.listen(5)
    print("Sensors server connected")

    while True:
        client, addr = server_socket.accept()
        print(f"Connection from {addr}")
        client_handler = threading.Thread(target=handle_client, args=(client,))
        client_handler.start()

# Command line execution
if __name__ == '__main__':
    sensors_setup()

```


5. Propellor schematics

